

ANSWERS & EXPLANATIONS

GENERAL STUDIES (P) TEST – 3787 (2023)

Q 1.C

- **Red soil develops on crystalline igneous rocks in areas of low rainfall in the southern and eastern part of the Deccan Plateau.** A long stretch of area is occupied by red loamy soil, along the piedmont zone of Western Ghat. **Yellow and red soils are also found in parts of Chhattisgarh and Odisha and in the southern parts of the middle Ganga plain.**
- **Due to a wide diffusion of iron in crystalline and metamorphic rocks, the soil develops a reddish color. When it occurs in a hydrated form, it looks yellow. The fine-grained red and yellow soils are normally fertile, whereas in dry upland areas the soil has poor fertility due to coarse-grained structure.** They are generally poor in humus, phosphorus, and nitrogen.
- **Hence option (c) is the correct answer.**

Q 2.C

- **Recent Context: 478 Khelo India centres have been approved in 375 districts of 29 states and union territories.**
- The Khelo India programme has been introduced to revive the sports culture in India at the grass-root level by building a strong framework for all sports played in our country and establish India as a great sporting nation.
- **It is a central sector scheme of Ministry of Sports and Youth Affairs.**
- **It aims to set up 1000 Khelo India centres across all districts of the country.**
- Talented players identified in priority sports disciplines at various levels by the High-Powered Committee will be provided annual financial assistance of INR 5 lakh per annum for 8 years.

- Following major **Indigenous/traditional games** have been identified:

Mallakhamb, MP	• It is a combination of Yoga, Gymnastics, and Martial Arts. • Many types of Mallakhamb as Pole, Rope, Hanging, Niradhar (without support), on cane, on floating platform, Mallakhamb with weapons etc.
Kalaripayattu, Kerala	• It is a form of Marital Art. • Has different techniques like meipayattu (physical body exercise), vadipayattu (fight using sticks), valpayattu (fight using swords) etc.
Gatka, Punjab	• Traditional martial art form, that is historically associated with the Sikh gurus. • Stick fighting between two or more practitioners, with wooden sticks (called Soti) intended to simulate swords.
Thang ta, Manipur	• Traditional martial art also called HUYEN LALLONG ("method of safe-guarding").
Yogasana	• Describes a seated posture in yoga typically used for meditation, such as a basic cross-legged position. • Examples: Padmasana (lotus pose), Siddhasana (perfect or accomplished pose), Vajrasana (thunderbolt pose) etc.
Silambam, TN	• Weapon-based Indian martial art, but also traditionally practised by the Tamil community of Sri Lanka and Malaysia.

Q 3.B

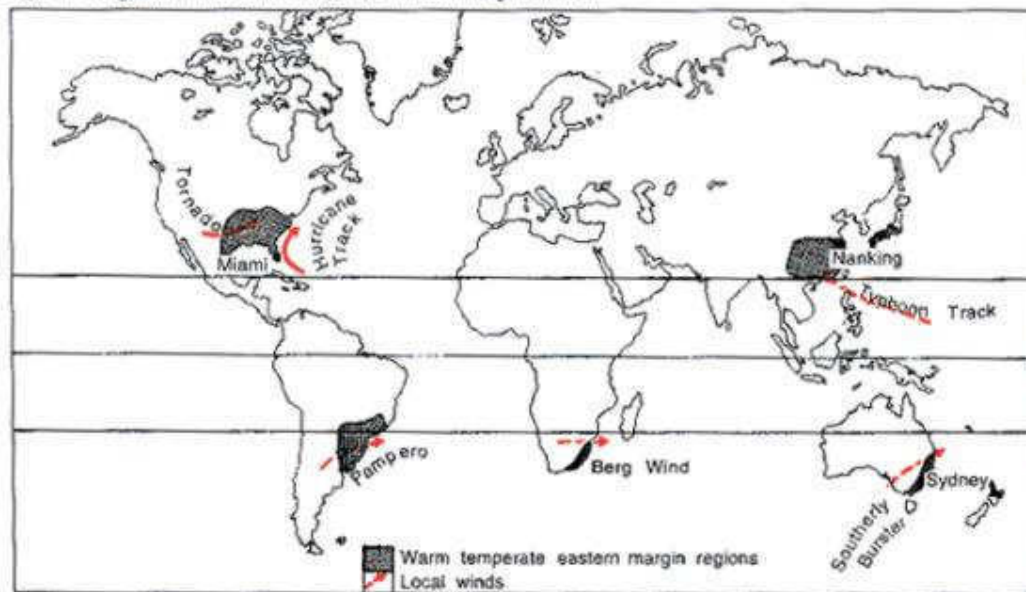
- The black soils are generally clayey, deep and impermeable. They **swell and become sticky when wet and shrink when dried**. So, during the dry season, these soils develop wide cracks. Thus, there occurs a kind of 'self ploughing'. Because of this character of slow absorption and loss of moisture, the black soil retains the moisture for a very long time, which helps the crops, especially, the rain-fed ones, to sustain even during the dry season. **Hence statement 2 is correct.**

- **Alluvial soils and not black soils are widespread in the northern plains and the river valleys. These soils cover about 40 per cent of the total area of the country.** Black soil covers most of the Deccan Plateau which includes parts of Maharashtra, Madhya Pradesh, Gujarat, Andhra Pradesh and some parts of Tamil Nadu. **Hence statement 1 is not correct.**
- Chemically, the black soils are rich in lime, iron, magnesia and alumina. They also contain potash. But they lack in phosphorous, nitrogen and organic matter. The colour of the soil ranges from deep black to grey.

Q 4.C

- **Features of Warm Temperate Eastern Margin Climate (China Type):**
 - **Influenced by the on-shore Trade Winds all the year-round, without any monsoon variations. Hence statement 1 is correct.**
 - **Fairly uniform distribution of rainfall throughout the year.** There is rain every month, except in the interior of central China, where there is a distinct dry season. Rain comes either from convectional sources or as orographic rain in summer, or from depressions in prolonged showers in winter. **Hence statement 2 is correct.**
 - It is sometimes referred to as the Gulf type or Natal type of climate.
 - This type of climate is found on the eastern margins of continents in warm temperate latitudes, just outside the tropics as seen clearly in the below figure:

Fig. 144 Regions with a Warm Temperate Eastern Margin Climate



Q 5.A

- Normally, temperature decreases with increase in elevation. It is called the normal lapse rate. At times, the situation is reversed and the normal lapse rate is inverted. It is called Inversion of temperature. Inversion is usually of short duration but quite common nonetheless.
- A long winter night with clear skies and still air is an ideal situation for inversion. The heat of the day is radiated off during the night, and by early morning hours, the earth is cooler than the air above. Over polar areas, a temperature inversion is normal throughout the year.
- **Statement 1 is correct:** Surface inversion promotes stability in the lower layers of the atmosphere. Smoke and dust particles get collected beneath the inversion layer and spread horizontally to fill the lower strata of the atmosphere. **Dense fogs in mornings are common occurrences, especially during the winter season. This inversion commonly lasts for a few hours until the sun comes up and beings to warm the earth.**
- **Statement 2 is not correct:** The inversion takes place in hills and mountains due to air drainage. Cold air at the hills and mountains, produced during night, flows under the influence of gravity. Being heavy and dense, the cold air acts almost like water and moves down the slope to pile up deeply in pockets and valley bottoms with warm air above. This is called air drainage. It protects plants from frost damages.

Q 6.C

- India with its area of 3.28 million sq. km accounts for 2.4 percent of the world's land surface area. It stands as the seventh largest country in the world. **Hence, statement 1 is not correct and statement 2 is correct.**
- The latitudinal and longitudinal extent of India is roughly about 30 degrees. The main land of India extends from 8° 4' N to 37° 17' N latitudes and from 68° 7' E to 97° 24' E longitudes. **Hence, statement 3 is not correct.**

Q 7.C

- The earth after being heated by insolation transmits the heat to the atmospheric layers near to the earth in a long waveform. The air in contact with the land gets heated slowly and the upper layers in contact with the lower layers also get heated. This process is called conduction.
- Conduction takes place when two bodies of unequal temperature are in contact with one another; there is a flow of energy from the warmer to a cooler body. The transfer of heat continues until both the bodies attain the same temperature or the contact is broken. Conduction is important in heating the lower layers of the atmosphere.
- **Statement 1 is correct:** The air in contact with the earth rises vertically on heating in the form of currents and further transmits the heat of the atmosphere. This process of vertical heating of the atmosphere is known as convection. **The convective transfer of energy is confined only to the troposphere.**
- **Statement 2 is correct:** The transfer of heat through the horizontal movement of air is called advection. The horizontal movement of the air is relatively more important than the vertical movement. **In middle latitudes, most of diurnal (day and night) variations in daily weather are caused by advection alone.** In tropical regions particularly in northern India during the summer season, local winds called 'loo' is the outcome of advection process.

Q 8.C

- **Statement 1 is correct:** The Indian Peninsula was **formed essentially by a great complex of very ancient gneisses and granites**, which constitutes a major part of it. Since the Cambrian period, the Peninsula has been standing like a rigid block with the exception of some of its western coast which is submerged beneath the sea and some other parts changed due to tectonic activity without affecting the original basement. As a part of the Indo-Australian Plate, it has been subjected to various vertical movements and block faulting. The rift valleys of the Narmada, the Tapi and the Mahanadi, and the Satpura block mountains are some examples of it. The Peninsula mostly consists of relict and residual mountains like the Aravali hills, the Nallamala hills, the Javadi hills, the Veliconda hills, the Palkonda range and the Mahendragiri hills, etc. The river valleys here are shallow with low gradients.
- **Statement 2 is correct:** The northern boundary of the Peninsular Block may be taken as an irregular line running from Kachchh along the western flank of the Aravali Range near Delhi and then roughly parallel to the Yamuna and the Ganga as far as the Rajmahal Hills and the Ganga delta. Apart from these, **the Karbi Anglong and the Meghalaya Plateau in the northeast and Rajasthan in the west are also extensions of this block.** The northeastern parts are separated by the Malda fault in West Bengal from the Chhotanagpur plateau. In Rajasthan, the desert and other desert-like features overlay this block.

Q 9.D

- Ocean currents are like river flow in oceans. They represent a regular volume of water in a definite path and direction. Ocean currents are influenced by two types of forces namely
 - primary forces that initiate the movement of water
 - secondary forces that influence the currents to flow.
- The primary forces that influence the currents are:
 - heating by solar energy;
 - wind;
 - gravity;
 - coriolis force
- Heating by solar energy causes the water to expand. That is why, near the equator the ocean water is about 8 cm higher in level than in the middle latitudes. This causes a very slight gradient and water tends to flow down the slope.
- **Wind blowing on the surface of the ocean pushes the water to move. Friction between the wind and the water surface affects the movement of the water body in its course.**

- Gravity tends to pull the water down to pile and create gradient variation.
- **The Coriolis force intervenes and causes the water to move to the right in the northern hemisphere and to the left in the southern hemisphere. These large accumulations of water and the flow around them are called Gyres. These produce large circular currents in all the ocean basins.**
- **Differences in water density affect vertical mobility of ocean currents. Water with high salinity is denser than water with low salinity and in the same way cold water is denser than warm water. Denser water tends to sink, while relatively lighter water tends to rise.**
- Cold-water ocean currents occur when the cold water at the poles sinks and slowly moves towards the equator. Warm-water currents travel out from the equator along the surface, flowing towards the poles to replace the sinking cold water.
- The topography and shape of ocean basins and nearby landmasses also influence ocean currents.
- **Hence option (d) is the correct answer.**

Q 10.D

- Many factors affects decomposition of organic matter.
- Moisture: **Maximum decomposition occurs at the intermediate level of moisture in the soil.** Soil gas exchange will be limited in high moisture soil leading to low oxygen concentrations and potentially anaerobic conditions. Also, low moisture content makes the lack of water thus limiting the microbial metabolism. Hence, both high and low moisture (dry) conditions inhibit the decomposition process.
- **Oxygen:** It accelerates the process of decomposition. Since major organisms depend on oxygen for respiration and reproduction, they can not multiply in hypoxic conditions i.e. lack of enough oxygen levels.
- **Composition:** Decomposition will be faster if the detritus is rich in nitrogen and water-soluble substances like sugars and it will be slower if the detritus is rich in lignin and chitin.
- **Temperature: Cool temperatures inhibit the decomposition and piling up of the organic content** in the soil takes place. Warm temperature makes the microorganisms active and faster will be the decomposition. **Hence option (d) is the correct answer.**
- **The soil pH can also influence plant growth** by its effect on the activity of beneficial microorganisms. Bacteria (like Rhizobia) that decompose soil organic matter are hindered in strong acid soils.

Q 11.A

- Peninsular India is drained by five major river systems: Mahanadi, Godavari, Krishna, Penner and Cauvery. Roughly triangular in shape with its base coinciding with the southern edge of the great plain of North India. The apex of the triangular plateau is at Kanyakumari, Tamil Nadu.
- Most of the peninsular **rivers flow west to east indicating it's general slope gradient of the west to east.** Narmada-Tapti is the **exceptions** which flow from east to west in a **rift caused by the divergent boundary.** The Peninsular Plateau is one of the oldest and stable landforms of the earth.
- The geology and climate of Peninsular India are two overriding physical controls affecting the rivers of the region. The rocks of the peninsular plateau of India extend further north-east beyond the Rajmahal hills and form a rectangular block known as the Meghalaya or the Shillong plateau. This **plateau has been separated from the main block** of the peninsular plateau by a wide gap known as the Garo-Rajmahal Gap. This gap was formed by down-faulting and was later on filled by sediments deposited by the Ganga.
- **Hence statement 1 is not correct and statement 2 is correct.**

Q 12.C

- **Peneplain** is an erosional plain carved by the agents of erosion. Rivers, rain, ice, and wind help to smooth out the irregularities of the earth's surface. They are almost plain hence called a peneplain.
- **Pediplain:** It is also an erosional plain. Mechanical weathering in arid and semi-arid areas wears back the mountain slopes to leave gently sloping pediments or pediplains, but some steep hills remain which are called inselbergs.
- **Structural plains:** These are structurally depressed areas of the world that make up some of the most extensive natural lowlands on the earth's surface. They are formed by horizontally bedded rocks, relatively undisturbed by the crustal movements of the earth. They include such great plains as the Russian Platform, the Great plains of the U.S.A., and the central lowlands of Australia. **Hence option (c) is the correct answer.**

- **Coastal plains:** These are depositional plains. In coastal regions, waves and winds often drive beach materials, mud, sand, or shingle landwards and deposit them on the coastal plain to form marine swamps, mud-flats, tidal and estuarine lowlands.

Q 13.A

- Water erosion in India takes place mainly in the form of sheet and gully erosion.
- Sheet erosion takes place on level lands after a heavy shower and the soil removal is not easily noticeable. But it is harmful since it removes the finer and more fertile topsoil. **Hence, statements 1 and 2 are correct.**
- Gully erosion is common on steep slopes. Gullies deepen with rainfall, cut the agricultural lands into small fragments and make them unfit for cultivation. A region with a large number of deep gullies or ravines is called a badland topography. **Hence, statement 3 is not correct.**

Q 14.A

- The word 'sedimentary' is derived from the Latin word sediment, which means settling. Rocks(igneous, sedimentary and metamorphic) of the earth's surface are exposed to denudational agents and are broken up into various sizes of fragments. Such fragments are transported by different exogenous agencies and deposited. **These deposits through compaction turn into rocks. This process is called lithification.**
- In many sedimentary rocks, the layers of deposits retain their characteristics even after lithification. Hence, we see a number of layers of varying thickness in sedimentary rocks like sandstone, shale etc.
- **Hence option (a) is the correct answer.**

Q 15.B

- Winter season in India extends roughly from November to March. **Winter Season is characterized by the following weather phenomenon:**
 - **Development of high pressure in the region lying to the north of the Himalayas. Hence, statement 2 is correct.** This centre of high pressure gives rise to the flow of air at the low level from the north towards the Indian subcontinent, south of the mountain range. The surface winds blowing out of the high pressure centre over Central Asia reach India in the form of a dry continental air mass.
 - All of Western and Central Asia remains under the influence of westerly jet streams winds along the altitude of 9-13 km from west to east.
 - Southern branch of the jet stream exercises an important influence on the winter weather in India.
 - The western cyclonic disturbances enter the Indian subcontinent from the west and the northwest during the winter months.
 - Tropical cyclones originate over the Bay of Bengal and the Indian ocean.
 - **An easterly jet stream flows over the southern part of the Peninsula in Summer month of June. Hence, statement 1 is not correct.**

Q 16.B

- Jet Streams and Upper Air Circulation during the summer season: The Easterly Jetstream is a unique and dominant feature of the northern hemispheric summer over southern Asia and northern Africa. The Easterly Jetstream is found near between 5° and 20°N. It flows from east to west over peninsular India at 6 – 9 km and over the Northern African region. It results in the reversal of upper air circulation patterns [High pressure switches to low pressure] and leads to the quick onset of monsoons.
- **Statement 2 is correct:** The easterly jet stream steers the tropical depressions into India. These depressions play a significant role in the distribution of monsoon rainfall over the Indian subcontinent. The tracks of these depressions are the areas of the highest rainfall in India.
- **Statement 1 is not correct:** An easterly jet stream flows over the southern part of the Peninsula in June and has a maximum speed of 90 km per hour. In August, it is confined to 15°N latitude, and in September up to 22° N latitudes. The easterlies normally do not extend to the north of 30° N latitude in the upper atmosphere. Hence, the influence of the easterly jet stream keeps increasing towards the north during the southwest monsoon period.

Q 17.A

- Social forestry means the management and protection of forests and afforestation on barren lands with the purpose of helping in environmental, social, and rural development.
- The National Commission on Agriculture(1976) has classified social forestry into three categories. These are Urban forestry, Rural forestry and Farm forestry:

- **Urban forestry** pertains to the raising and management of trees on public and privately owned lands in and around urban centres such as green belts, parks, roadside avenues, industrial and commercial green belts, etc. **Hence pair 3 is correctly matched.**
- **Rural forestry** lays emphasis on the promotion of agro-forestry and community-forestry.
- **Agro-forestry** is the raising of trees and agriculture crops on the same land inclusive of the waste patches. **Hence pair 1 is correctly matched.**
- **Community forestry** involves the raising of trees on public or community land such as the village pasture and temple land, roadside, canal bank, strips along railway lines, and schools, etc. It aims at providing benefits to the community as a whole.
- **Farm forestry** means the process under which farmers grow trees for commercial and non-commercial purposes on their farmlands. **Hence pair 2 is correctly matched.**

Q 18.D

- **Since tides are caused by the earth-moon-sun positions** which are known accurately, the tides can be predicted well in advance. This helps the navigators and fishermen plan their activities. Tidal flows are of great importance in navigation.
- **Importance of tides:**
 - Fish may concentrate during ebb tides. Commercial fishermen follow the tides and learn to fish during levels of highest concentration to improve their economic investment and to make more efficient use of their time.
 - Tides affect other aspects of oceanic life, including the reproductive activities of fish and ocean plants. Floating plants and animals ride the tidal currents between the breeding areas and deeper waters. **The tides help remove pollutants and circulate nutrients ocean plants and animals need to survive.**
 - Crabs, mussels, snails, seaweed, and other edible sea life inhabit the tidal zone. Small tide pools may also contain small fish and sea vegetables. The sea life found in these regions are often harvested for food. Without the regular washing of the tides, these complex and abundant creatures would die and food resources would diminish.
 - Tides affect the depth and currents in and around coastal areas. Ships may need to navigate the waters during high tide in some areas or risk running aground. Pilots take into consideration the water level, width of channels and direction of the water flow to determine the best time to travel.
 - **Two high tides and two low tides occur during every 24-hour period.** The predictability of the tides, fast movement of water during the inflow and outflow can provide a source of renewable energy to communities living along the coast.
- **Hence option (d) is the correct answer.**

Q 19.A

- **Recently, Allahabad University (AU) alumnus and a renowned physicist working as emeritus faculty at the Indian Institute of Science, Education and Research (IISER), Pune, Deepak Dhar has been selected for the prestigious Boltzmann Medal for 2022—the first Indian to ever be awarded this prestigious honour. Hence, statement 2 is not correct.**
- The Boltzmann Medal, has been instituted by Commission on Statistical Physics (C3) of the International Union of Pure and Applied Physics (IUPAP).
- The award consists of the gilded Boltzmann medal with the inscription of Austrian physicist and philosopher Ludwig Boltzmann.
- **It is given out once in three years to honour outstanding achievements in statistical physics. Hence, statement 1 is correct.**

Q 20.C

- **Statement 1 is correct:** The largest part of the northern plain is formed of older alluvium. They lie above the flood plains of the rivers and present a terrace like feature. This part is known as **bhangar**. The soil in this region **contains calcareous deposits locally known as kankar.**
- **Statement 2 is correct:** The newer, younger deposits of the flood plains are called **khadar**. They are **renewed almost every year and so are fertile, thus, ideal for intensive agriculture.**

Q 21.C

- **Stalactites hang as icicles of different diameters (from the ceiling of underground caves).** Normally they are broad at their bases and taper towards the free ends showing up in a variety of forms.

- **Stalagmites rise up from the floor of the caves. Stalagmites form due to dripping water from the surface or through the thin pipe**, of the stalactite, immediately below it. Stalagmites may take the shape of a column, a disc, with either a smooth, rounded bulging end or a miniature crater-like depression. The stalagmite and stalactites eventually fuse to give rise to columns and pillars of different diameters.
- **Hence both the statements are correct.**

Q 22.B

- **Recently, the Gujarat forest department is planning to use SIMBA or Software with Intelligent Marking Based identification of Asiatic lions** which is a photo-identification software, specifically designed to distinguish patterns or marks. The software has been developed by Hyderabad-based “teliolabs”. The software is trained with sufficient training data and is able to isolate the region of interest (ROI) and separate them for further identification. **Hence, statement 3 is correct.**
- Asiatic Lions are listed as **Endangered** on the IUCN red list of species. **Hence, statement 1 is not correct.**
- **Asiatic lions living in Gir and other protected areas of the Saurashtra region of Gujarat** are the world’s only wild population of lions outside Africa. **Hence, statement 2 is correct.**

Q 23.C

- **Recently, India has decided to build Nuclear Power Plants In "Fleet Mode" From 2023. Under the fleet mode, a nuclear power plant is expected to be built over a period of five years from the first pour of concrete.**
 - Currently, India operates 22 reactors with a total capacity of 6780 MW in operation.
- **Since 2005, when India signed a nuclear agreement with the United States (US), the country has made nuclear cooperation with other countries a significant aspect of its diplomatic initiatives. At present, India has civil nuclear agreements with 14 countries that vary in letter and spirit.**
- India’s civil nuclear agreements have been crucial in according credibility to its status as a responsible nuclear power. To articulate the future of India’s civil nuclear energy cooperation, it is important to recognize the exceptional record that India holds in the global civil nuclear domain, given its status as a non-signatory to the Treaty on the Non-Proliferation of Nuclear Weapons (or the Non-Proliferation Treaty, NPT) and its non-membership in the Nuclear Suppliers Group (NSG).
- There are 14 countries with which India has forged such agreements: Argentina, **Australia, Canada**, Czech Republic, France, **Japan**, Kazakhstan, Mongolia, Namibia, Russia, South Korea, United Kingdom, US and **Vietnam**.
- **India’s civil nuclear accords aim to develop mutually beneficial economic, scientific and technical cooperation for the peaceful uses of nuclear energy.**
 - A central tenet of these agreements is the focus on the “peaceful purposes” of nuclear power; this entails the “use of information, nuclear material, equipment or components in such fields as research, power generation, medicine, agriculture and industry.
- Currently, there is no collaboration between India and Israel on nuclear energy.
- **Hence, option (c) is the correct answer.**

Q 24.C

- The word El-Nino means 'Child Christ', because this current appears around Christmas in December. December is a summer month in Peru (Southern Hemisphere) and in almost every three to seven years, this sets in warm currents off the coast of Peru in the eastern Pacific and affects the weather in many places including India.
- **El-Nino is merely an extension of the warm equatorial current which gets replaced temporarily by cold Peruvian current or Humbolt current. Hence statement 1 is correct.**
- This current increases the temperature of water on the Peruvian coast by 10 degree Celcius. This results in
 - the distortion of equatorial atmospheric circulation.
 - irregularities in the evaporation of sea water.
 - reduction in the amount of planktons which further reduces the number of fish in the sea.
- **El-Nino is used in India for forecasting long range monsoon rainfall. The El-Nino event has led to delayed or total failure of monsoon over most parts of the country in 1990-91, 1998, 2006-07 and 2015-16. Hence statement 2 is correct.**

Q 25.B

- **The Kuroshio Current begins off the east coast of Philippines and Taiwan and flows northeastward past Japan, where it merges with the easterly drift of the North Pacific Current. Hence statement 1 is not correct.**
- An ocean gyre is a large system of circular ocean currents formed by global wind patterns and forces created by Earth's rotation. The movement of the world's major ocean gyres helps drive the ocean conveyor belt. The ocean conveyor belt circulates ocean water around the entire planet.
- The Kuroshio Current is the warm western boundary current of the North Pacific's subtropical gyre. **Hence statement 2 is correct.**
- It is analogous to the Gulf Stream in the Atlantic Ocean. Its warm waters are carried polewards as the North Pacific Drift keeping the ports of Alaskan coast ice-free in winter.

Q 26.B

- The Himalayan region contains many 'Passes', which are a stretch of land, usually passing through the mountains, that connects two different regions. A few of the important passes of the Indian subcontinent are:
- **Zoji La:** Zoji La is a high mountain pass state of Jammu and Kashmir, located between Srinagar and Leh in the Western Himalayan mountain range. It separates Kashmir Valley to its west from Dras Valley to its northeast.
- **Bara Lacha La:** It is a mountain pass in Zaskar range, connecting Lahaul district in Himachal Pradesh to Ladakh in Jammu and Kashmir.
- **Shipki La:** It is a mountain pass on the India-China border. It is located in Kinnaur district in the state of Himachal Pradesh. The river Sutlej enters India (from Tibet) near this pass. The pass is one of India's border posts for trade with Tibet along with Nathu La in Sikkim, and Lipulekh in Uttarakhand.
- **Jelep La:** It is a mountain pass between Sikkim (India) and Tibet (China). It is on a route that connects Lhasa to India.
- **Nathu La:** It is a mountain pass in the Himalayas in East Sikkim district. It connects Sikkim with Tibet (China). The pass forms a part of an offshoot of the ancient Silk Road.
- **Bolan Pass:** The Bolan Pass is a mountain pass through the Toba Kakar Range of Balochistan province in western Pakistan. The pass is an 89 km stretch of the Bolan river valley from Rindli in the south to Kolpur in the north. It is made up of a number of narrow gorges and stretches.
- **Khyber Pass:** The Khyber Pass is a mountain pass in the north of Pakistan, close to the border with Afghanistan. It traverses through the Spin Ghar mountains. An integral part of the ancient Silk Road, it has long had substantial cultural, economic, and geopolitical significance for Eurasian trade. Throughout history, it has been an important trade route between Central Asia and the Indian subcontinent. Among the Muslim invasions of the Indian subcontinent, the famous invaders coming through the Khyber Pass are Mahmud Ghaznavi, and the Afghan Muhammad Ghor and the Turkic-Mongols.
- **Hence option (b) is the correct answer.**

Q 27.B

- **Hydrological Drought:** It results when the availability of water in different storages and reservoirs like aquifers, lakes, reservoirs, etc. falls below what the precipitation can replenish. **Hence option (b) is the correct answer.**
- **Meteorological Drought:** It is a situation when there is a prolonged period of inadequate rainfall marked with mal-distribution of the same over time and space.
- **Agricultural Drought:** It is also known as soil moisture drought, characterised by low soil moisture that is necessary to support the crops, thereby resulting in crop failures. Moreover, if an area has more than 30 per cent of its gross cropped area under irrigation, the area is excluded from the drought-prone category.
- **Ecological Drought:** When the productivity of a natural ecosystem fails due to the shortage of water and as a consequence of ecological distress, damages are induced in the ecosystem.

Q 28.B

- Extra-tropical cyclone, also known as wave cyclone or mid-latitude cyclone, is a type of storm system formed in middle or high latitudes, in regions of large horizontal temperature variations called frontal zones.
- They develop when a wave forms on a frontal surface separating a warm air-mass from a cold air mass. Thus, extra-tropical cyclones form along the polar front. In the northern hemisphere, warm air blows from the south and cold air from the north of the front.

CURRENT

CURRENT SHOTS 365 YEARLY MAGAZINE



CAPITAL ACCOUNT CONVERTIBILITY [CAC]

Freedom to Convert Domestic
Currency for Capital
Account Transactions

MYSURU DECLARATION ON SERVICE DELIVERY BY PANCHAYATS WAS SIGNED

GATI SHAKTI

A digital platform to bring 16
Ministries together for integrated
planning and coordinated
implementation of infrastructure
connectivity projects

KEN-BETWA RIVER INTERLINKING PROJECT

FIRST PROJECT UNDER
THE NATIONAL
PERSPECTIVE PLAN

JUDIMA RICE WINE A HOME-MADE RICE WINE OF ASSAM'S DIMASA TRIBE BAGGED GI TAG

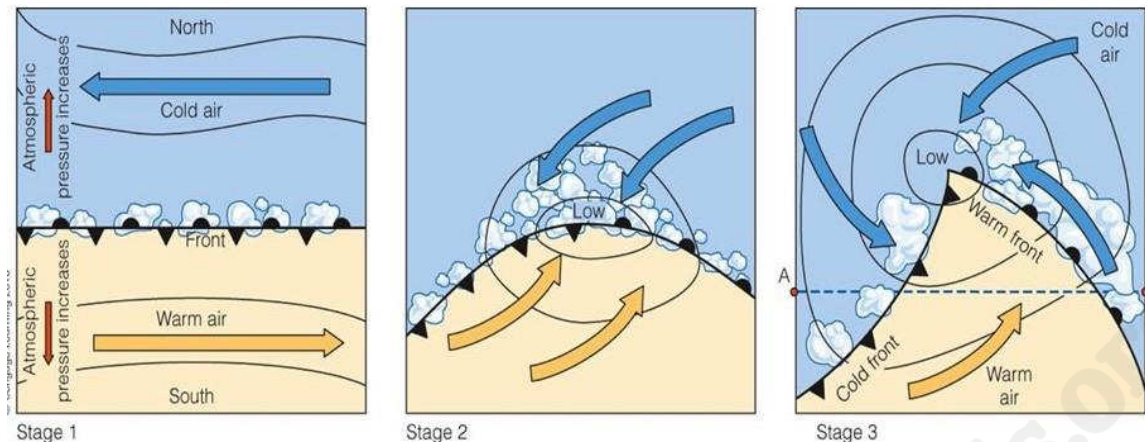
THAMIRABARANI CIVILIZATION

This is the oldest civilization
perhaps, older than the Vaigai
civilization which is believed to
be 2,600 years old

A Comprehensive Current Affairs Revision Magazine for Civil
Services - EPFO - RRB - SSC - NDA - CDS and All Other
Competitive Exams.

- When the pressure drops along the front, the warm air moves northwards and the cold air move towards the south setting in motion an anti-clockwise cyclonic circulation. The cyclonic circulation leads to a well-developed extra-tropical cyclone, with a warm front and a cold front.
- **They cover a larger area and can originate over the land and sea. The wind velocity in a tropical cyclone is much higher and it is more destructive. The extra-tropical cyclones move from west to east in the northern hemisphere. Hence statement 1 is not correct and statement 2 is correct.**

(a) The genesis and early development of an extratropical cyclone in the Northern Hemisphere



Q 29.C

- The rotation of the earth about its axis affects the direction of the wind. This force is called the Coriolis force after the French physicist who described it in 1844. It deflects the wind in the right direction in the northern hemisphere and to the left in the southern hemisphere. The deflection is more when the wind velocity is high. The Coriolis force is directly proportional to the angle of latitude. **It is maximum at the poles and is absent at the equator. Hence statement 2 is correct.**
- The Coriolis force acts perpendicular to the pressure gradient force. The pressure gradient force is perpendicular to an isobar (a line connecting points having the same atmospheric pressure). The higher the pressure gradient force, the more is the velocity of the wind and the larger is the deflection in the direction of the wind. As a result of these two forces operating perpendicular to each other, in the low-pressure areas the wind blows around it. **At the equator, the Coriolis force is zero and the wind blows perpendicular to the isobars. The low pressure gets filled instead of getting intensified. That is the reason why tropical cyclones are not formed near the equator. Hence, statement 1 is correct.**
- The velocity and direction of the wind are the net results of the wind generating forces. The winds in the upper atmosphere, 2 - 3 km above the surface, are free from frictional effect of the surface and are controlled by the pressure gradient and the Coriolis force. When isobars are straight and when there is no friction, the pressure gradient force is balanced by the Coriolis force and the resultant wind blows parallel to the isobar. This wind is known as the geostrophic wind.

Q 30.A

- The Deccan plateau is bounded by the Eastern and Western Ghats (in East and West respectively).
- **Western Ghats:**
 - They are comparatively higher in elevation and more continuous than the Eastern Ghats.
 - The height increases from north to south.
 - They are locally known by different names such as Sahyadri in Maharashtra, Nilgiri hills in Karnataka and Tamil Nadu, and Anaimalai hills and Cardamom hills in Kerala.
 - Most of the Peninsular rivers have their origin in the Western Ghats.
- **Eastern Ghats:**
 - They are lower in height and are discontinuous.
 - They are highly eroded by the rivers such as the Mahanadi, the Godavari, the Krishna, the Kaveri, etc which form their deltas in the region.

- Some of the important ranges include the Javadi hills, the Palconda range, the Nallamala hills, the Mahendragiri hills, etc.
- **Eastern and the Western Ghats meet at Nilgiri Hills.**
- **Hence option (a) is the correct answer.**

Q 31.B

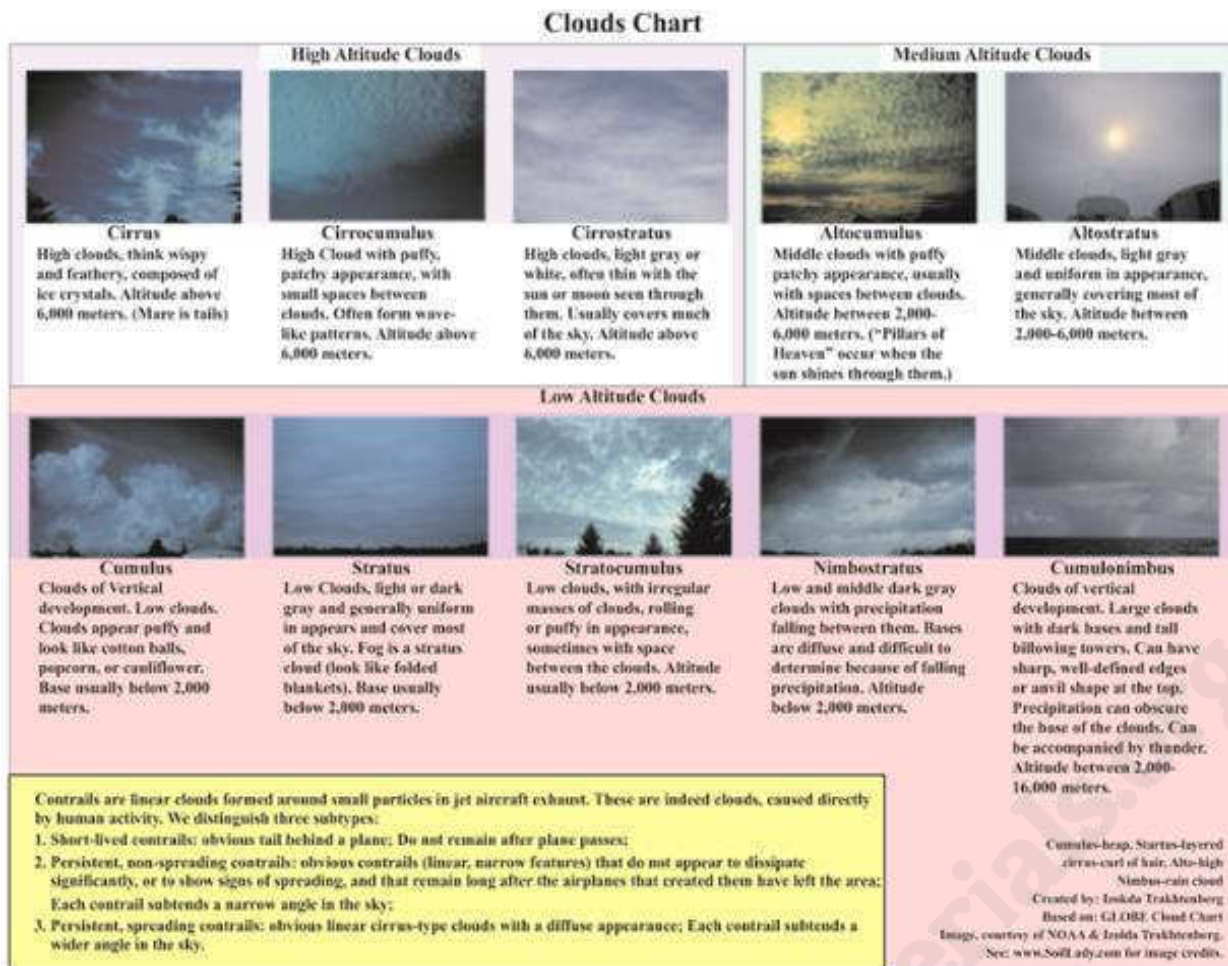
- **Mojave desert** is an arid rain-shadow desert and the driest desert in North America. It is in the Southwestern United States, primarily within southeastern California and southern Nevada. It extends from the Sierra Nevada range to the Colorado Plateau and merges with the Great Basin to the north and the Sonoran Desert to the south. **Hence option (b) is the correct answer.**
- **Patagonian desert** is also known as the Patagonian Steppe is the largest desert in Argentina. It is located primarily in Argentina and is bounded by the Andes, to its west, and the Atlantic Ocean to its east, in the region of Patagonia, southern Argentina. To the north the desert grades into the Cuyo Region and the Pampas. The central parts of the steppe are dominated by shrubby and herbaceous plant species albeit to the west, where precipitation is higher, bushes are replaced by grasses. Topographically the deserts consist of alternating tablelands and massifs dissected by river valleys and canyons. The more western parts of the steppe host lakes of glacial origin and grades into barren mountains or cold temperate forests along valleys.
- **Gibson desert** is the arid zone in the interior of Western Australia. Hence, pair 3 is not correctly matched. The desert lies south of the Tropic of Capricorn between the Great Sandy Desert (north), the **Great Victoria Desert** (south). It is known for its undulating red dunes, desert pavement, and sandy plains and the desert grass Triodia. It is a fire-prone landscape.

Q 32.D

- The continents that the Equator passes through include **Australia and Oceania, South America, Africa and Asia**. This imaginary line that circles the world traverses 13 countries:
 - Kiribati in Australia and Oceania;
 - Ecuador, Colombia and Brazil in South America;
 - Sao Tome and Principe, Gabon, Democratic Republic of Congo, Congo, Kenya, Uganda and Somalia in Africa and
 - Maldives and Indonesia in Asia.

Q 33.B

- **Cumulus:** This is a vertical cloud with a rounded top and horizontal base, typical of humid tropical regions, associated with up-rising convectional currents. Its great white globular masses may look grey against the sun but it is a fair-weather cloud. **Hence option (b) is correct.**
- **Cirrus:** This looks fibrous and appears like wisps in the blue sky; it is often called 'mares' tails'. It indicates fair weather and often gives a brilliant sunset.
- **Nimbostratus:** This is a dark, dull cloud, clearly layered, and is also known as a 'rain cloud'. It brings continuous rain, snow or sleet.
- **Cumulonimbus:** This is, in fact, an overgrown cumulus cloud, extending for a tremendous vertical height from a base of 2,000 feet to over 30,000 feet. Its black and white globular masses take a fantastic range of shapes. Its cauliflower top often spreads out like an anvil. This is frequently seen in tropical afternoons. It is also referred to as a 'thunder-cloud' and brings convectional rain, accompanied by lightning and thunder.



Q 34.C

- **Statement 1 is correct:** The Himalayas consist of three parallel ranges-Great Himalayas or Inner Himalayas also known as the Himadri because it remains covered with snow throughout the year.
- **Statement 2 is correct:** This range includes the world's highest peak, Mt. Everest (8848 m), Annapurna (8,078) in Nepal, Nanga Parbat (8,126 m) in Kashmir and also the highest peak in India, Kanchenjunga (8,598).
- Number of glaciers descend from the Inner Himalayas such as the Gangotri and the Yamunotri. They lie in the Uttarakhand. These glaciers are the perennial source of water for the Ganga and Yamuna rivers, respectively.

Q 35.B

- Tropic of Cancer is an imaginary line, at an angle of 23.5 degrees North from the Equator, that passes through roughly the middle of India. If you run this imaginary line around Earth, it passes through 17 countries, India being one of them.
- The Tropic of Cancer passes through **eight states** in India: Gujarat (Jasdan), Rajasthan (Kalinjarh), Madhya Pradesh (Shajapur), Chhattisgarh (Sonhat), Jharkhand (Lohardaga), West Bengal (Krishnanagar), Tripura (Udaipur) and Mizoram (Champhai) in that order. **Hence option (b) is the correct answer.**
- Mahi River is the only river in India that cuts the Tropic of Cancer twice, first in Madhya Pradesh from where it flows towards Rajasthan and enters Gujarat where it cuts for the second time. Udaipur in Tripura is the city nearest to the Tropic of Cancer.

Q 36.D

- **Recently, while speaking at the United Nations Security Council meeting on the Ukraine crisis, India's Permanent Representative to the UN called on all nations to ensure full implementation of the Biological and Toxin Weapons Convention.**
- Formally known as "The Convention on the Prohibition of the Development, Production and Stockpiling of Bacteriological (Biological) and Toxin Weapons and on their Destruction", **the Convention was**

negotiated by the Conference of the Committee on Disarmament in Geneva, Switzerland. It opened for signature on 10 April 1972 and entered into force on 26 March 1975. Hence, statement 3 is correct.

- The BWC supplements the 1925 Geneva Protocol, which had prohibited only the use of biological weapons.
- The Biological Weapons Convention (BWC) effectively prohibits the development, production, acquisition, transfer, stockpiling and use of biological and toxin weapons. **Hence, statement 1 is correct.**
- It was the first multilateral disarmament treaty banning an entire category of weapons of mass destruction (WMD). **Hence, statement 2 is correct.**

Q 37.D

- **Recently, NTPC awarded India's first Green Hydrogenbased Microgrid Project at Simhadri, Andhra Pradesh. Hence statement 2 is not correct.** It would be a precursor to large scale hydrogen energy storage projects. The project offers benefits of decarbonising far-off regions (e.g. Ladakh) and meet the clean energy goals because of its features like sustainability, easy to store etc.
- Hydrogen is the most abundant chemical element in the universe. **Despite its abundance, hydrogen does not naturally exist as a gas in usable quantities, occurring almost entirely in compounds, such as water.** Therefore, hydrogen must be produced using industrial methods. Most of these involve the reforming of natural gas – a fossil fuel.
- **Green hydrogen is produced by splitting water into hydrogen and oxygen using an electrolyzer powered by renewable energy sources such as wind and solar. Hence statement 1 is not correct.**

Q 38.A

- **Option 1 is correct:** Solifluction involves slow downslope flowing soil mass or fine grained rock debris saturated or lubricated with water.
- **Option 2 is not correct:** Earthflow is a rapid movement of water-saturated clayey or silty earth materials down low-angle terraces or hillsides.
- **Option 3 is not correct:** Slump is a type of landslide which is relatively rapid and perceptible movement. The materials involved are relatively dry. It is slipping of one or several units of rock debris with a backward rotation with respect to the slope over which the movement takes place.

Q 39.B

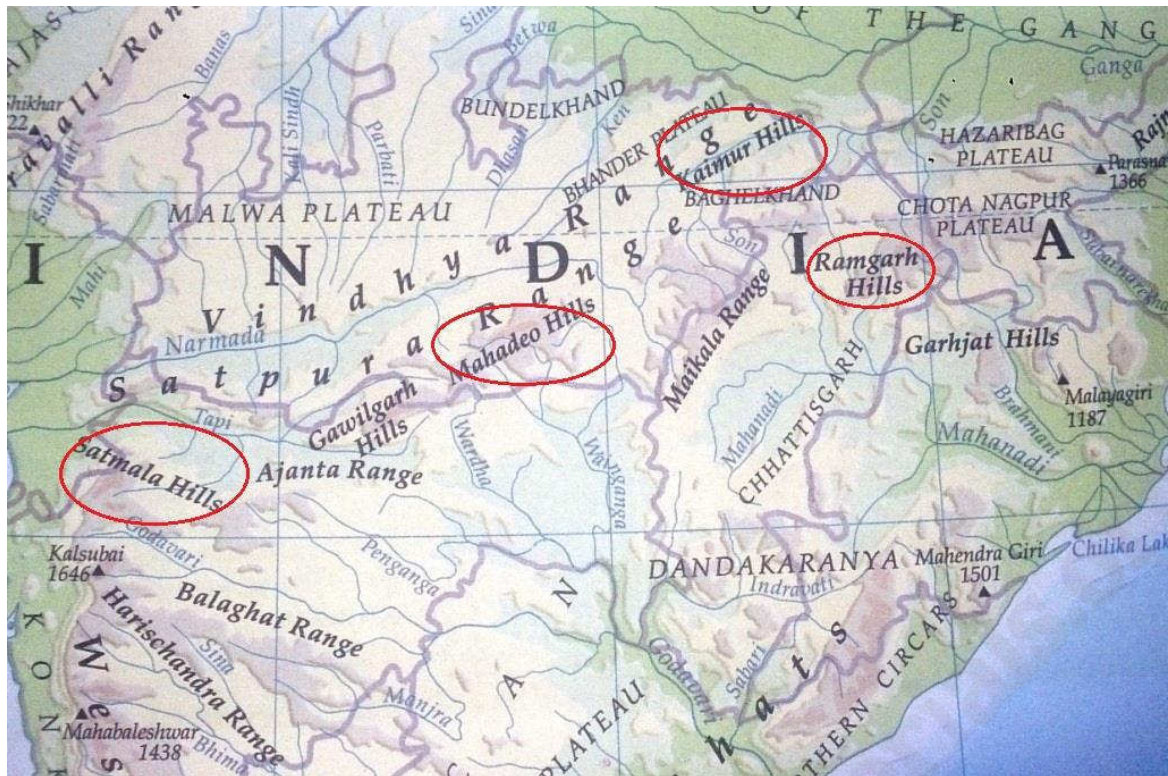
- **Statement 1 is not correct:** The maximum temperature of the oceans is always at their surfaces because they directly receive the heat from the sun and the heat is transmitted to the lower sections of the oceans through the process of conduction. It results in the decrease of temperature with the increasing depth, but the rate of decrease is not uniform throughout. Temperature structure of oceans can be described as three layer system. First layer (Top layer) is about 500m thick with temperatures ranging between 20 to 25 °C
- Second layer (Thermocline) is characterised by rapid decrease in temperature with increasing depth. It is about 500m-1000m thick. Third layer is very cold and extends upto deep ocean floor. Here temperature change with the depth is very slight.
- **Statement 2 is correct:** The average temperature of surface water of the oceans is about 27°C and it gradually decreases from the equator towards the poles. The oceans in the northern hemisphere record relatively higher temperature than in the southern hemisphere. The highest temperature is not recorded at the equator but slightly towards north of it. The average annual temperatures for the northern and southern hemisphere are around 19° C and 16° C respectively. This variation is due to the unequal distribution of land and water in the northern and southern hemispheres.

Q 40.C

- The lofty Himalayas in the north along with its extensions act as an effective climatic divide. The towering mountain chain provides an invincible shield to protect the subcontinent from the cold northern winds. These cold and chilly winds originate near the Arctic circle and blow across central and eastern Asia. **Hence, statement 1 is correct.**
- The Himalayas also trap the monsoon winds, forcing them to shed their moisture within the subcontinent. **Hence, statement 2 is correct.**

Q 41.B

- The peninsular region consists of many important mountain ranges like Aravalli, Satmala, Ajanta, Mahadeo, Maikal, Kaimur, Garhjat etc.
- Their relative order is: Satmala – Mahadeo – Kaimur - Ramgarh
- **Hence, option (b) is the correct answer.**

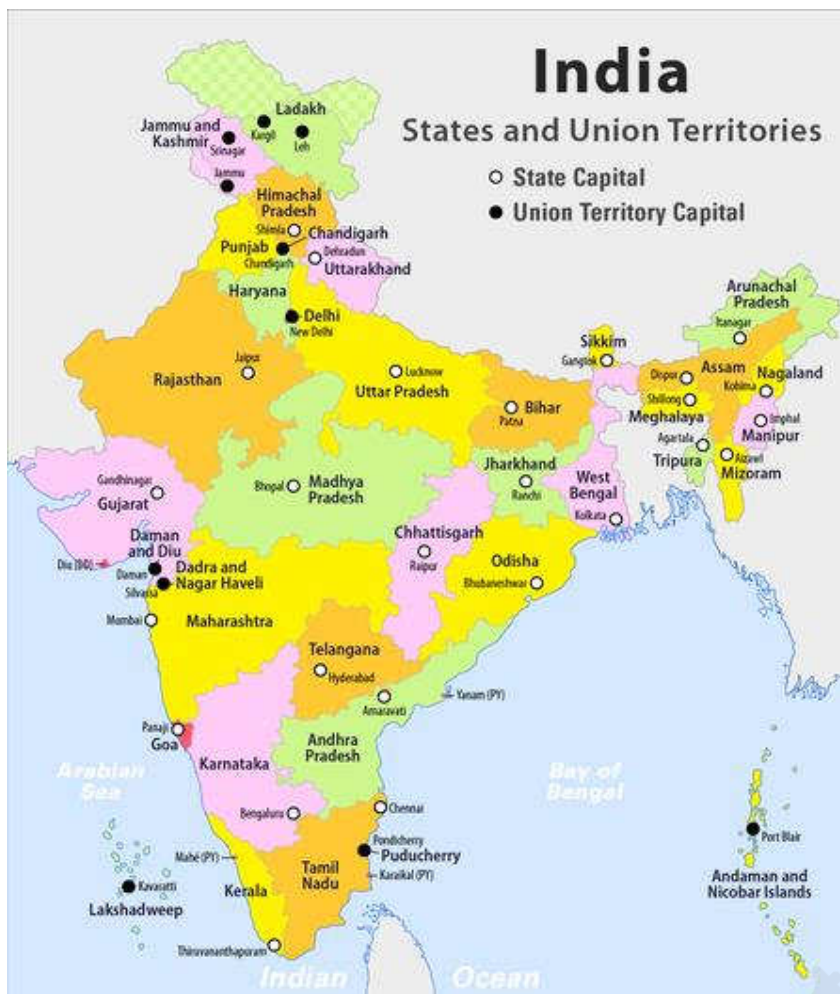


Q 42.A

- Near the equator, the tropics receive the most rain on a consistent basis. As a result, the freshwater falling into the ocean helps decrease the salinity of the surface water in that region. Hence salinity is lower than the average 35 ppt (parts per thousand) in equatorial waters.
- **The lower salinity water rests above the higher salinity dense water. Salinity, generally, increases with depth and there is a distinct zone called the halocline, where salinity increases sharply.** Other factors being constant, increasing the salinity of seawater causes its density to increase. High salinity seawater, generally, sinks below the lower salinity water. This leads to stratification by salinity. **Hence statement 2 is not correct.**
- The average salinity of the Indian Ocean is 35ppt. The low salinity trend is observed in the Bay of Bengal due to the influx of river water by the river Ganga. On the contrary, **the Arabian Sea shows higher salinity due to high evaporation and a low influx of freshwater. Hence statement 1 is correct.**

Q 43.C

- **Jaipur** is the capital and the largest city of the Indian state of Rajasthan. It is situated at 75.8°E longitudes.
- **Bhopal** is the capital city of the Indian state of Madhya Pradesh. It is situated at 77°25'E longitudes.
- **Lucknow** is the capital city of the Indian state of Uttar Pradesh. It is situated at 80°57'E longitudes.
- **Raipur** is the capital city of the Indian state of Chhattisgarh. It is situated at 81.63°E longitudes.



- Hence option (c) is the correct answer.

Q 44.A

- **Temperate Continental (Steppe) Climate:**
 - A steppe is a dry, grassy plain. Steppes occur in temperate climates, which lie between the tropics and polar regions. Temperate regions have distinct seasonal temperature changes, with cold winters and warm summers. Though they lie in the Westerly wind belt, they are so remote from the maritime influence that the grasslands are practically treeless. In the northern hemisphere, the grasslands are far more extensive and are entirely continental. **Hence statement 1 is correct.**
 - In Eurasia, they are called the Steppes, and stretch eastwards from the shores of the Black Sea across the great Russian plain to the foothills of the Altai Mountains, a distance of well over 2,000 miles.
 - In the southern hemisphere, there is a maritime influence in the steppe type of climate. Its annual precipitation is always more than the average 20 inches because of the warm ocean currents that wash the shores of the steppe-lands here. Pretoria, in South Africa, has an annual precipitation of 26 inches with the wettest months in November, December, January and February, the summer season of the southern hemisphere. There are three months (June, July and August) without any rain. This is the period of drought. The dry season is particularly pronounced in temperate grasslands adjoining deserts, for example in Australia. **Hence statement 2 is not correct.**

Q 45.C

- The atmosphere is composed of gases, water vapour and dust particles. The proportion of gases changes in the higher layers of the atmosphere in such a way that oxygen will be almost in negligible quantity at the height of 120 km.
- Similarly, carbon dioxide and water vapour are found only up to 90 km from the surface of the earth. Ozone is another important component of the atmosphere found between 10 and 50 km above the earth's surface and acts as a filter and absorbs the ultra-violet rays radiating from the sun and prevents them from reaching the surface of the earth. **Hence statement 1 is correct.**

Permanent Gases of the Atmosphere		
Constituent	Formula	Percentage by Volume
Nitrogen	N ₂	78.08
Oxygen	O ₂	20.95
Argon	Ar	0.93
Carbon dioxide	CO ₂	0.036
Neon	Ne	0.002
Helium	He	0.0005
Krypton	Kr	0.001
Xenon	Xe	0.00009
Hydrogen	H ₂	0.00005

- An inert gas is a gas that has extremely low reactivity with other substances. The noble gases—helium, argon, neon, xenon, krypton, radon are examples of inert gases. **Hence statement 2 is correct.**

Q 46.C

- Molasses are thick associations of continental and marine clastic sedimentary rocks that consist mainly of sandstones and shales formed as shore deposits. The depositional environments involved include beaches, lagoons, river channels, and backwater swamps.
- Mizoram is also known as the 'Molasses basin'. It is made up of soft unconsolidated deposits. **Hence option (c) is the correct answer.**

Q 47.D

- The following countries share their borders with the Mediterranean Sea: **Spain**, France, Monaco, Italy, Malta, Slovenia, Croatia, Bosnia and Herzegovina, Montenegro, Albania, Greece, Turkey, **Cyprus**, **Syria**, Lebanon, Israel, Egypt, Libya, Tunisia, Algeria, and Morocco. **Hence option (d) is the correct answer.**



Q 48.A

- Formerly known as “Lagrange,” Vigil Mission is a European Space Agency's upcoming space weather mission that will monitor our active and unpredictable Sun to protect vital infrastructure on Earth, in space as well as space explorers from its violent outbursts.
- Hence, statement 1 is correct.**
- The mission will give us advance warning of oncoming solar storms and therefore more time to protect spacecraft in orbit, infrastructure on the ground and explorers now and in the future, unshielded by Earth's magnetic field and vulnerable to our star's violent outbursts like solar flares and ‘coronal mass ejections. **Hence, statement 2 is not correct.**
- Vigil Mission will be launched in the mid-2020s and will be placed in the 5th Lagrange point orbit.
 - Lagrange Points are positions in space where the gravitational forces of a two-body system like the Sun and the Earth produce enhanced regions of attraction and repulsion. These can be used by spacecraft to reduce fuel consumption needed to remain in position.

Q 49.C

- **Supply of semiconductors, which plummeted due to Covid-related disruptions but had started picking up as manufacturing chains normalised, is now being threatened once again by the Ukraine crisis.** This is on account of supply of two key raw materials, neon and palladium, that are at a risk of being constrained.
- A semiconductor is a material product usually comprised of silicon, which conducts electricity more than an insulator, such as glass, **but less than a pure conductor, such as copper or aluminum. Hence, statement 1 is not correct.**
- Their conductivity and other properties can be altered with the introduction of impurities, called **doping**, to meet the specific needs of the electronic component in which it resides. **Hence, statement 2 is correct.**
- Broadly speaking, semiconductors fall into four main product categories:
 - **Memory chips** serve as temporary storehouses of data and pass information to and from computer devices' brains. The consolidation of the memory market continues, driving memory prices so low that only a few giants like Toshiba, Samsung, and NEC can afford to stay in the game.
 - **Microprocessors:** These are central processing units that contain the basic logic to perform tasks. Intel's domination of the microprocessor segment has forced nearly every other competitor, with the exception of Advanced Micro Devices, out of the mainstream market and into smaller niches or different segments altogether.
 - **Commodity Integrated Circuit:** Sometimes called "standard chips", these are produced in huge batches for routine processing purposes. Dominated by very large Asian chip manufacturers, this segment offers razor-thin profit margins that only the biggest semiconductor companies can compete for.
 - **Complex SOC: "System on a Chip"** is essentially all about the creation of an integrated circuit chip with an entire system's capability on it. The market revolves around the growing demand for consumer products that combine new features and lower prices. With the doors to the memory, microprocessor, and commodity integrated circuit markets tightly shut, the SOC segment is arguably the only one left with enough opportunity to attract a wide range of companies. **Hence, statement 3 is correct.**

Q 50.A

- **Option 1 is correct.** A convergent boundary is where the crust is destroyed as one plate dives under another. The location where the sinking of a plate occurs is called a subduction zone. There are three ways in which convergence can occur. These are: (i) between an oceanic and continental plate; (ii) between two oceanic plates; and (iii) between two continental plates.
- **Option 2 is correct.** A divergent boundary is where a new crust is created (generated) as the plates pull away from each other. The sites where the plates move away from each other are called spreading sites.
- **Option 3 is not correct.** A transform boundary is where the crust is neither produced nor destroyed as the plates slide horizontally past each other. Transform faults are the planes of separation generally perpendicular to the mid-oceanic ridges.

Q 51.A

- The Inter Tropical Convergence Zone (ITCZ) is a low pressure zone located at the equator where trade winds converge, and so, it is a zone where air tends to ascend. **Hence, statement 1 is correct.**
- The location of the ITCZ gradually varies with the seasons.
- In July, the ITCZ is located around 20°N-25°N latitudes (over the Gangetic plain), sometimes called the monsoon trough. This monsoon trough encourages the development of thermal low over north and northwest India.
- Due to the shift of ITCZ, the trade winds of the southern hemisphere cross the equator between 40° and 60°E longitudes and start blowing from southwest to northeast due to the Coriolis force. It becomes southwest monsoon.
- In winter, the ITCZ moves southward, and so the reversal of winds from northeast to south and southwest, takes place. They are called northeast monsoons. **Hence, statement 2 is not correct.**
- As the axis of the monsoon trough oscillates, there are fluctuations in the track and direction of these depressions, and the intensity and the amount of rainfall vary from year to year.

Q 52.A

- **Recently, Turkey has decided to implement an international convention on naval passage through two of its strategic straits (Bosphorus and Dardanelles), which would allow them to limit the movement of Russian warships between the Mediterranean Sea and the Black Sea.**

- The Bosphorus and Dardanelles straits, also known as the Turkish Straits or the Black Sea Straits, connect the Aegean Sea and the Black Sea via the Sea of Marmara. It is the only passage through which the Black Sea ports can access the Mediterranean and beyond.
- **Bosphorus is a natural strait connecting the Black Sea to the Sea of Marmara**, thus being a very strategic waterway. It was a river in the valley during the Tertiary period of the continents, which was drowned by the sea at the end of this period. Its length is 32 kilometers (20 miles) in the north to south direction, width varies between 730-3300 meters (800-3600 yards), and depth is between 30-120 meters (100-395 feet). **Bosphorus strait separates the European part from the Asian part of Istanbul.** The surface current flows always from north to south; however, a strong counter current under the surface creates swirls and eddies.



- Hence, option (a) is the correct answer.

Q 53.D

- The Brahmaputra, one of the largest rivers of the world, has its **origin in the Chemayungdung glacier of the Kailash range near the Mansarovar lake**. From here, it traverses eastward longitudinally for a distance of nearly 1,200 km in a dry and flat region of southern Tibet, where it is known as the Tsangpo, which means 'the purifier.' The Rango Tsangpo is the major right-bank tributary of this river in Tibet. It emerges as a turbulent and dynamic river after carving out a deep gorge in the Central Himalayas near Namcha Barwa (7,755 m).
- **The river emerges from the foothills under the name of Siang or Dihang.** It enters India from Arunachal Pradesh. Flowing southwest, it receives its main left-bank tributaries, viz., Dibang or Sikang and Lohit; thereafter, it is known as the Brahmaputra.
- The Brahmaputra receives numerous tributaries in its 750 km long journey through the Assam valley. Its major left bank tributaries are the Burhi Dihing and Dhansari (South) whereas the **important right bank tributaries** are the Subansiri, Kameng, **Manas** and Sankosh. The Subansiri which has its origin in Tibet, is an antecedent river. The Brahmaputra enters into Bangladesh near Dhubri and flows southward. In Bangladesh, the Tista joins it on its right bank from where the river is known as the Jamuna. **It finally merges with the river Padma, which falls in the Bay of Bengal.**
- Hence option (d) is the correct answer.

Q 54.C

- India is flanked by the Indian Ocean on three sides in the south and girdled by a high and continuous mountain-wall in the north.
- **As compared to the landmass, water heats up or cools down slowly.** This differential heating of land and sea creates different air pressure zones in different seasons in and around the Indian subcontinent.
- **The difference in air pressure causes a reversal in the direction of monsoon winds.**
- Hence both the statements are correct.

Q 55.C

- The Kashmir Himalayas are famous for Karewa formations. Karewas are the thick deposits of glacial clay (and not alluvial deposits) and other materials embedded with moraines. The Karewas are also useful for the cultivation of Zafran, which is a local variety of saffron.
- **Hence option (c) is the correct answer.**

Q 56.B

- Equatorial type of climate is associated with all the given features given below: The most outstanding feature of the equatorial climate is its great uniformity of temperature throughout the year.
- There is no winter.
- Cloudiness and heavy precipitation help to moderate the daily temperature, so that even at the equator itself, the climate is not unbearable. In addition, regular land and sea breezes assist in maintaining a truly equable climate.
- The diurnal range of temperature is small, and so is the annual range. High temperature and abundant rainfall in the equatorial regions support a luxuriant type of vegetation i.e. the Tropical Rain Forests.
- The equatorial vegetation comprises a multitude of evergreen trees that yield tropical hardwood, e.g. mahogany, ebony, etc. It supports a luxuriant type of vegetation – the tropical rain forest. Amazon tropical rain forest is known as Selvas. Trees of single species are very scarce in such vegetation.
- There are smaller palm trees, climbing plants like the lianas or rattan which may be hundreds of feet long and epiphytic and parasitic plants that live on other plants. Under the trees grow a wide variety of ferns, orchids and lalang. The tallest trees attain a height of over 150 feet.
- **Hence option (b) is the correct answer.**

Q 57.B

- **The Achanakmar-Amarkantak Biosphere Reserve** is the most dramatic and ecologically diverse landscape in the Chhattisgarh and Madhya Pradesh states of India. The area of this Biosphere Reserve is considered to be one of the major watersheds of peninsular India. It separates the rivers that drain into the Arabian Sea and the Bay of Bengal. The reserve is also the source of three major river systems: the Narmada, the Johilla and the Son River. Maikal hill ranges together with Vindhya and Satpura lie within the Achanakmar-Amarkantak Biosphere Reserve. UNESCO designated it a biosphere reserve in 2012. **Hence option (b) is the correct answer.**
- **Similipal National Park:** It is a national park and a tiger reserve in the Mayurbhanj district in the Indian state of Odisha. It is part of the Mayurbhanj Elephant Reserve, which includes three protected areas — Similipal Tiger Reserve, Hadgarh Wildlife Sanctuary and Kuldiha Wildlife Sanctuary. Simlipal National Park derives its name from the abundance of red silk cotton trees growing in the area. The park is home to Bengal tiger, Asian elephant, gaur, and chausingha, along with some of the beautiful waterfalls like Joranda and Barehipani Falls. This protected area is part of the UNESCO World Network of Biosphere Reserves since 2009.
- **The Agasthyamalai Biosphere Reserve** was Established in 2001 and includes 3,500.36 km² of which 1828 km² is in Kerala and 1672.36 km² is in Tamil Nadu. Agasthyamalai Biosphere Reserve became part of World Network of Biosphere Reserves in 2016. Is also under UNESCO's world list of biosphere reserve.
- **The Pachmarhi Biosphere Reserve** is located in the Satpura Range of Madhya Pradesh. The Satpura range forms the watershed between the river Narmada to the north and river Tapi to the south. UNESCO designated it a biosphere reserve in 2009. It includes three wildlife conservation units: Bori Sanctuary, Pachmarhi Sanctuary and Satpura National Park.

Q 58.B

- Tillite is the sedimentary rock formed out of deposits of glaciers. The Gondwana system of sediments from India is known to have its counterparts in six different landmasses of the Southern Hemisphere. At the base, the system has thick tillite indicating extensive and prolonged glaciation. Counterparts of this succession are found in Africa, Falkland Island Madagascar, Antarctica and Australia besides India.
- The overall resemblance of the Gondwana-type sediments clearly demonstrates that these landmasses had remarkably similar histories. The glacial tillite provides unambiguous evidence of palaeoclimates and also of drifting of continents. **Hence, (b) is the correct answer.**
- Pumice is a very light and porous volcanic rock formed when a gas-rich froth of glassy lava solidifies rapidly. Hence, it is an igneous rock. Marble is a metamorphic rock composed of recrystallized carbonate minerals, most commonly calcite or dolomite. Quartzite is a metamorphic rock formed when quartz-rich sandstone or chert has been exposed to high temperatures and pressures.

Q 59.B

- The flow of water through well-defined channels is known as 'drainage' and the network of such channels is called a 'drainage system'.
- The drainage pattern of an area is the outcome of the geological time period, nature and structure of rocks, topography, slope, amount of water flowing and the periodicity of the flow. Some of the different types of drainage patterns are:
- **Dendritic:** The drainage pattern resembling the branches of a tree is known as Dendritic. It is formed in areas of gently sloping land. Himalayan rivers follow a dendritic pattern. **Hence pair 1 is not correctly matched.**
- **Radial:** When the rivers originate from a hill and flow in all directions, the drainage pattern is known as Radial. The rivers originating from the Amarkantak range present a good example of it.
- **Trellis:** When the primary tributaries of rivers flow parallel to each other and secondary tributaries join them at right angles, the pattern is known as Trellis. Rivers in the upper part of the Himalayan region form a trellis pattern. **Hence pair 3 is correctly matched.**
- **Centripetal:** When the rivers discharge their waters from all directions in a lake or depression, the pattern is known as the Centripetal pattern. For example, Loktak lake in Manipur. **Hence pair 2 is correctly matched.**

Q 60.D

- The crust is the outer thin layer with a total thickness normally between 30-50 km. **Hence, statement 1 is not correct.**
- The continents are composed of lighter silicates; silica and aluminum also called 'sial' and they are made up of Granitic rocks while the oceans have the heavier silicates; silica and magnesium also called 'sima' and are made up of basaltic rocks. **Hence both statements 2 and 3 are also not correct.**

Q 61.B

- The actual amount of water vapour present in the air, which is expressed in grams per cubic metre, is called the absolute humidity. But more important from the point of view of weather studies is the relative humidity. This is the ratio between the actual amount of water vapour and the total amount the air can hold at a given temperature, expressed as a percentage.
- Warm air can hold more water vapour than cold air, so if it contains only half the amount it could carry, the relative humidity is 50 per cent. In the equatorial regions, over 80 per cent is common in the morning, 'which means the air contains four-fifths as much water vapour as it can carry. **When the relative humidity reaches 100 per cent, the air is completely saturated. Hence option (b) is the correct answer.**
- The air temperature is said to be at dew-point. **Further cooling will condense the water vapour into clouds or rain.** It is thus clear that when relative humidity is high the air is moist, as in the equatorial regions; when it is low, the air is dry as in the deserts. For clouds to form and rain to start, the air does have to reach 100% relative humidity, but only where the clouds are forming or where the rain is coming from.

Q 62.D

- **Recently, the United Nations Environment Programme (UNEP) identified five major contributors to the global development, transmission and spread of antimicrobial resistance (AMR) in a new report.**
- The major sources of AMR mentioned in the UNEP report released on the sidelines of the fifth session of the UN Environment Assembly include: **Poor sanitation, sewage and waste effluent; Effluent and waste from pharmaceuticals manufacturing; Effluent and waste from healthcare facilities; Use of antimicrobial and manure in crop production; Releases, effluent and waste in animal production.** **Hence, option (d) is the correct answer.**
- Pollution from antimicrobial agents disrupts the microbial composition of environmental media and affects biodiversity and ecosystem services. Water, soil and air then serve as vehicles for spreading antimicrobial resistant microbes and affects human, animal and other environmental reservoirs.

Q 63.D

- All observations from telescopes reveal that the planetary bodies, the Sun, Moon, satellites and stars have circular outlines from whichever angle they are seen. They are strictly spheres. Earth, by analogy, cannot be the only exception. **Hence, statement 1 is correct.**

- The distant horizon viewed from the deck of a ship at sea, or from a cliff on land always and everywhere is circular in shape. This circular horizon widens with increasing altitude and could only be seen on a spherical body.
- The sun sets and rises at different times at different places. As the earth rotates from west to east, places in east see the sun earlier than those in the west. If the earth were flat, the whole world would have sunrise and sunset at the same time. **Hence, statement 2 is correct.**
- During the lunar eclipse, the shadow cast by the earth on the moon is always circular. It takes the outline of an arc of a circle. Only an sphere can cast such a circular shadow. **Hence, statement 3 is correct.**

Q 64.B

- **Statement 1 is not correct:** Meander is not a landform but is only a type of channel pattern. In large flood and delta plains, rivers rarely flow in straight courses. Loop-like channel patterns called meanders develop over flood and delta plains. These are associated with rivers **in old stage**.
- **Statement 2 is correct:** Following factors play important role in the formation of meanders: (i) propensity of water flowing over very gentle gradients to work laterally on the banks (ii) unconsolidated nature of alluvial deposits making up the banks with many irregularities which can be used by water exerting pressure laterally; (iii) Coriolis force acting on the fluid water deflecting it like it deflects the wind.
- **Meanders are formed by River Jhelum due to the local base level of sediments** provided by a primeval lake that existed millions of years ago in Kashmir Valley. Rivers generally form meanders in their mature stage but Jhelum, flowing through the Kashmir valley, forms meander in its youth stage.

Q 65.B

- **Recently, the Brahmaputra River got connected with the Ganga river via the India and Bangladesh Inland Water Transit and Trade Protocol Route.** The vessel started its sail from Patna on National Waterway-1 (river Ganga) and passed through Bhagalpur, Manihari, Sahibganj, Farakka, Tribeni, Kolkata, Haldia, Hemnagar; Indo Bangladesh Protocol (IBP) route through Khulna, Narayanganj, Sirajganj, Chilmari and National Waterway-2 through Dhubri, and Jogighopa covering 2,350 km.
- The people's Republic of Bangladesh and the Republic of India have a long-standing and time-tested Protocol on Transit and Trade through inland waterways of both countries. **This Protocol was first signed in 1972** (immediately after the independence of Bangladesh). It was last renewed in 2015 for five years with a provision for its automatic renewal for a further period of five years giving long term assurance to various stakeholders.
- Under the protocol, Inland water vessels of one country can transit through **specific routes** of the other country. **Hence, statement 1 is not correct.**
- **It is under the overall ambit of the Inland Waterway Authority of India. Hence, statement 2 is correct.**
- Under the Second Addendum on Protocol on Inland Water Transit and Trade between India and Bangladesh, 2020, Inland vessels of both the countries can ply on the designated protocol route and dock at **Ports of Call** in each country, notified for loading/unloading of cargo.

Q 66.A

- In India, the mangrove forests are highly developed in the Andaman and Nicobar Islands and the Sunderbans of West Bengal. Other areas of significance are the Mahanadi, the Godavari and the Krishna deltas. As per the **Indian State of Forest Report 2021**, mangrove cover in the states are as follows :
- Andhra Pradesh - 405 sq km
- Gujarat - 1175 sq km
- Maharashtra - 324 sq km
- Odisha - 259 sq km
- West Bengal - 2114 sq km
- Andaman & Nicobar Islands - 616 sq km

Sl. No.	State/UT	Very Dense Mangrove	Moderately Dense Mangrove	Open Mangrove	Total	Change with respect to ISFR 2019
1.	Andhra Pradesh	0	213	192	405	1
2.	Goa	0	21	6	27	1
3.	Gujarat	0	169	1,006	1,175	-2
4.	Karnataka	0	2	11	13	3
5.	Kerala	0	5	4	9	0
6.	Maharashtra	0	90	234	324	4
7.	Odisha	81	94	84	259	8
8.	Tamil Nadu	1	27	17	45	0
9.	West Bengal	994	692	428	2,114	2
10.	A&N Islands	399	168	49	616	0
11.	D&NH and Daman & Diu	0	0	3	3	0
12.	Puducherry	0	0	2	2	0
Total		1,475	1,481	2,036	4,992	17

- Hence option (a) is the correct answer.

Q 67.B

- Recently, concerns have been raised regarding the impact of the Wild Life (Protection) Amendment Bill, 2021 on the Van Gujjars.
- The Van Gujjars, the nomadic pastoral community of the Uttarakhand Himalayas. They continue to practice seasonal migration across forests in Uttarakhand, Uttar Pradesh and Himachal Pradesh in pursuit of their pastoral livelihood. Hence option (d) is the correct answer.
- They have the right of a qafila (caravan) of Van Gujjars to migrate to their summer homesteads in the bugyals (Himalayan alpine meadows) located within the Govind Pashu Vihar National Park in Uttarkashi district.
- The rotational grazing of the Van Gujjars and their buffaloes also help maintain water sources through embankments, removal of invasive species, facilitation in seed dispersal, remanuring the grazing areas and paths and creating fire lines within forests to protect the ecosystem for other species too.
- In fact, Van Gujjars' activities such as lopping of leaves enable the sustenance of herbivores. It also fosters healthier growth and density of the lopped trees.

Q 68.A

- The continental shelf is a gently sloping and relatively flat extension of a continent that is covered by the oceans. Seaward, the shelf ends abruptly at the shelf break, the boundary that separates the shelf from the continental slope.
- Continental shelves contain valuable resources, such as oil and gas, and minerals. Oil and gas are formed from organic material that accumulates on the continental shelf. Over time the material is buried and transformed into oil and gas by heat and pressure. Hence statement 2 is correct.
- Continental shelves are home to the largest concentration of bottom-dwelling marine life, and they are the sites of major fishing grounds. Hence statement 1 is correct.
- Trenches are the deepest parts of the oceans. The trenches are relatively steep-sided, narrow basins. They are some 3-5 km deeper than the surrounding ocean floor. They occur at the bases of continental slopes and along island arcs and are associated with active volcanoes and strong earthquakes. Hence statement 3 is not correct.

Q 69.B

- Recently, the Stree Manoraksha project was launched during the Women's day celebrations.
- It was launched by the National Institute of Mental Health and Neuro - Science (NIMHANS) and the Union Minister for Women and Child Development. Hence, statement 1 is not correct.
- The project is aimed at extending mental health training to 6000 One Stop Centre functionaries across India. Hence, statement 2 is correct.

Q 70.D

- **The Himachal and Uttarakhand Himalayas:** This part lies approximately between the Ravi in the west and the Kali (a tributary of Ghaghara) in the east. It is drained by two major river systems of India, i.e. the Indus and the Ganga. All three ranges of the Himalayas are prominent in this section and these are : the Great Himalayan range, the Lesser Himalayas (which is locally known as Dhauladhar in Himachal Pradesh and Nagtibha in Uttarakhand) and the Shiwalik range from the North to the South. **The two distinguishing features of this region from the point of view of physiography are the 'Shiwalik' and 'Dun formations'. Duns are longitudinal valleys. Some important duns located in this region are the Chandigarh-Kalka dun, Nalagarh dun, Dehra Dun, Harike dun and the Kota dun, etc. Dehra Dun is the largest of all the duns with an approximate length of 35-45 km and a width of 22-25 km.**

Q 71.C

- The "Bucharest Nine" is a group of nine NATO countries in Eastern Europe that became part of the US-led military alliance after the end of the Cold War.
- The Bucharest Nine or Bucharest Format, often abbreviated as the B9, was founded on November 4, 2015, and takes its name from Bucharest, the capital of Romania.
- The B9 are, apart from **Romania and Poland, Hungary, Bulgaria, the Czech Republic, Slovakia, and the three Baltic republics of Estonia, Latvia, and Lithuania.** All nine countries were once closely associated with the now dissolved Soviet Union, but later chose the path of democracy. Romania, Poland, Hungary, and Bulgaria are former signatories of the now dissolved Warsaw Pact military alliance led by the Soviet Union. Estonia, Latvia, and Lithuania were part of the former Union of Soviet Socialist Republics (USSR).
- All members of the B9 are part of the European Union (EU) and North Atlantic Treaty Organisation (NATO).
- **Hence, option (c) is the correct answer.**

Q 72.A

- India's climate is controlled by a number of factors which can be broadly divided into two groups
 - Factors related to location and relief,
 - Factors related to air pressure and winds.
- Latitude is one of the factors related to Location and Relief
- Tropic of Cancer passes through the central part of India in the east-west direction. Thus, the northern part of India lies in the sub-tropical and temperate zone and the part lying south of the Tropic of Cancer falls in the tropical zone. Hence statement 1 is correct. The tropical zone being nearer to the equator experiences high temperatures throughout the year with small daily and annual range. **Hence statement 1 is correct.**
- The area north of the Tropic of Cancer being away from the equator experiences an extreme climate with high daily and annual range of temperature.
- India lies wholly in the northern and eastern hemispheres. The mainland of India extends from 8°4' 28" N to 37°17' 53" N latitudes. Tropic of Cancer lies at 23°27' N of the terrestrial Equator. It is the circle marking the latitude 23.5 degrees north, where the sun is directly overhead at noon on June 21, the beginning of summer in the northern hemisphere. Beyond the Tropic of Cancer, Northern India does not observe overhead Sun on the summer solstice of the northern hemisphere. **Hence statement 2 is not correct.**

Q 73.A

- **Statement 1 is correct:** All planets orbit around the sun in a counter-clockwise direction.
- **Statement 2 is correct:** Earth has the highest density with density of 5.517 gm/cm³. Second highest is Mercury with density of 5.44 gm/cm³.
- **Statement 3 is not correct:** All planets, except Mercury and Venus, have atleast one natural satellite. Like Earth has one, Mars has two, Jupiter has 16 and so on.

Other Information about Solar system:

- Jupiter is the largest planet by size.
- All planets have density above density of water (1 gm/cm³), except Saturn (0.7 gm/cm³). So if Saturn is placed in a very large pool of water, it will float, while all other planets will sink.

Q 74.B

- **Statement 1 is not correct:** The solar output received at the top of the atmosphere varies slightly in a year due to the variations in the distance between the earth and the sun. During its revolution around the sun, the earth is farthest from the sun (152 million km) on 4th July. This position of the earth is called aphelion. On 3rd January, the earth is the nearest to the sun (147 million km). This position is called the perihelion. Therefore, the annual insolation received by the earth on 3rd January is slightly more than the amount received on 4th July. However, the effect of this variation in the solar output is masked by other factors like the distribution of land and sea and the atmospheric circulation.
- **Statement 2 is correct:** The insolation received at the surface varies from about 320 Watt/m² in the tropics to about 70 Watt/m² in the poles. Maximum insolation is received over the subtropical deserts, where the cloudiness is the least. The Equator receives comparatively less insolation than the tropics.
- Generally, at the same latitude, the insolation is more over the continent than over the oceans. In winter, the middle and higher latitudes receive less radiation than in summer.

Q 75.A

- **Foehn and Chinook** are the local winds of the Temperate Continental Climate. The Foehn in Switzerland and the Chinook on the eastern slopes of the Rockies have considerable effect on the local pastures. They are hot winds and may raise the temperature by about 4 to 5 degree Celcius, thus melting the snow in winter or early spring and is beneficial for agriculture.
- Warm moist air moves from the Mediterranean region and brings precipitation on the southern slopes of the Alps. This air becomes warm due to adiabatic rise in temperature, loses its original characteristics and descends on the northern slopes of the Alps in Switzerland, Germany and Austria. This dry warm air is called as Foehn. **Hence, the correct answer is option (a).**
- **Sirocco and Mistral** are the local winds of the Mediterranean Climate. Sirocco: This is a hot, dry dusty wind that originates in the Sahara Desert. It may occur at any time of the year but is most frequent in spring. It blows outwards in a southerly direction from the desert interiors into the cooler Mediterranean Sea. It is known by different names in different countries around Mediterranean. Mistral: It is cold wind from north, rushing down the Rhone valley in France over to the Mediterranean sea.

Q 76.C

- There are three main reasons for the excessive cold in north India during the winter season :
 - Continentality- States like Punjab, Haryana and Rajasthan being far away from the moderating influence of sea experience continental climate.
 - The snowfall in the nearby Himalayan ranges creates a cold wave situation.
 - Around February, the cold winds coming from the Caspian Sea and Turkmenistan bring cold wave along with frost and fog over the northwestern parts of India.
- The easterly jet stream sets in along 15°N latitude only after the western jet stream has withdrawn itself from the region i.e. around the month of June. This easterly jet stream is held responsible for the burst of the monsoon in India. It does not play a role in an excessive cold climate in North India during the winter season. **Hence options 1, 2 and 3 are correct.**

Q 77.C

- **Recently, the telecom companies are backing the proposal to regulate the Content Delivery Networks, that were kept out of Telecom Regulatory Authority of India's 2017 recommendations on net neutrality. Hence, statement 2 is correct.**
- A content delivery network (CDN) refers to a **geographically distributed group of servers** which work together to provide fast delivery of Internet content. **Hence, statement 1 is correct.**
- A CDN allows for the quick transfer of assets needed for loading Internet content including HTML pages, javascript files, stylesheets, images, and videos. The popularity of CDN services continues to grow, and today the majority of web traffic is served through CDNs, including traffic from major sites like Facebook, Netflix, and Amazon.
- **While a CDN does not host content** and can't replace the need for proper web hosting, **it does help cache content at the network edge, which improves website performance.** Many websites struggle to have their performance needs met by traditional hosting services, which is why they opt for CDNs.
- Bandwidth consumption costs for website hosting is a primary expense for websites. Through caching and other optimizations, **CDNs are able to reduce the amount of data an origin server must provide, thus reducing hosting costs for website owners.**

Q 78.D

- The gravitation force (g) is not the same at different latitudes on the surface. It is greater near the poles and less at the equator. This is because of the distance from the centre at the equator being greater than that at the poles.
- The gravity values also differ according to the mass of material. The uneven distribution of mass of material within the earth influences this value. The reading of the gravity at different places is influenced by many other factors. **These readings differ from the expected values. Such a difference is called gravity anomaly. Hence, option (d) is the correct answer.**

Q 79.D

- India has two major island groups: Andaman and Nicobar, and Lakshadweep. Lakshadweep (and Minicoy) islands are situated in the Arabian Sea off the Malabar coast. They are scattered between 8°N-12°N latitude and 71°E -74°E longitude. The entire group of islands is made of coral deposits.
- The entire group of islands is divided by the Eleventh-degree channel. **Hence only statement 1 is correct and statements 2 and 3 are not correct.**
- **Hence option (d) is the correct answer.**

Q 80.C

- **Pair 1 is not correctly matched:** Hamada is a rocky desert. This consists of large stretches of bare rocks, swept clear of sand and dust by the wind. The exposed rocks are thoroughly smoothed and polished. The region is bare and sterile.
- **Pair 2 is correctly matched:** Reg or stony desert. This is composed of extensive sheets of angular pebbles and gravels which the winds are not able to blow off. In Libya and Egypt the termserir is used; elsewhere in Africa, stony deserts are called reg.
- **Pair 3 is not correctly matched:** Erg or sandy desert. This is a sea of sand which typifies the popular idea of desert scenery. Winds deposit vast stretches of undulating sand dunes in the heart of the deserts. The intricate patterns of ripples on the dune surfaces indicate the direction of the winds.

Q 81.A

- **Statement 1 is correct: The continents that the Equator passes through include** Australia and Oceania, South America, Africa, and **Asia**. This imaginary line that circles the world traverses 13 countries:
 - Kiribati in Australia and Oceania;
 - Ecuador, Colombia, and Brazil in South America;
 - Sao Tome and Principe, Gabon, Democratic Republic of Congo, Congo, Kenya, Uganda, and Somalia in Africa and
 - **Maldives and Indonesia in Asia.**
- **Statement 2 is not correct: The Mahi river in India passes through the Tropic of Cancer twice.**
- **The Congo River in Africa passes through the equator twice.**

Q 82.B

- **Recently, the Union Law Minister stated that there was a need to adopt new, cutting edge technologies of Machine Learning (ML) and Artificial Intelligence (AI) to increase the efficiency of the justice delivery system.**
- **The 'Supreme Court Vidhik Anuvaad Software' short termed as 'SUVAS' is a machine assisted translation tool trained by Artificial Intelligence.** This Tool is especially designed for Judicial Domain and at present, has the capacity and capability of translating English Judicial documents, Orders or Judgments into nine vernacular languages scripts and vice versa. This is the first step towards introduction of Artificial Intelligence in Judicial Domain.
- A pet project of Chief Justice of India S A Bobde, **the Supreme Court Portal for Assistance in Court's Efficiency (SUPACE)** is a tool that collects relevant facts and laws and makes them available to a judge.
- **Hence, option (b) is the correct answer.**

Q 83.D

- Darjeeling and Sikkim Himalayas are famous for Duar formations. Duars are alluvial floodplains that lie south of the outer foothills of the Himalayas and north of the Brahmaputra River basin. **Hence statement 1 is correct.**
- This region is divided by the Sankosh River into Eastern and Western Duars. **Hence statement 2 is correct.**

- The Western Duars are also known as the Bengal Duars, and the Eastern Duars also as the Assam Duars. Duars is analogous to the Terai in northern India and southern Nepal.
- Duars play an important role in the region as they support the tea plantations. Significantly, the Shiwaliks are absent in this region. **Hence statement 3 is correct.**

Q 84.C

- A catchment is an area of land where water collects when it rains, often bounded by hills. As the water flows over the landscape it finds its way into streams and down into the soil, eventually feeding the river. Some of this water stays underground and continues to slowly feed the river in times of low rainfall.

S No	River Basin	CA (Sq. Km)	Major river	River Length, km
1	Indus (Upto border)	321289	Indus(India)	1114
2	Ganga- Brahmaputra-Meghna			
a	Ganga	861452	Ganga	2525
b	Brahmaputra	194413	Brahmaputra (India)	916
c	Barak & others	41723	Barak	564
3	Godavari	312812	Godavari	1465
4	Krishna	258948	Krishna	1400
5	Cauvery	81155	Cauvery	800
6	Subernarekha	29169	Subernarekha	395
			Burhabalang	164
7	Brahmani & Baitarni	51822	Brahmani	799
			Baitarni	355
8	Mahanadi	141589	Mahanadi	851
9	Pennar	55213	Pennar	597
10	Mahi	34842	Mahi	583

- **Hence option (c) is the correct answer.**

Q 85.A

- Anticyclones are the opposite of cyclones, with high pressure in the centre and the isobars far apart. The pressure gradient is gentle and winds are light.
- **Statement 1 is correct:** Anti-cyclones normally herald fine weather. Skies are clear. The air is calm and temperatures are high in summer but cold in winter. In winter intense cooling of the lower atmosphere may result in thick fogs. Anti-cyclonic conditions may last for days or weeks and then fade out quietly.
- **Statement 2 is not correct:** Winds in anticyclone' blow outwards and are also subject to deflection, but they blow clockwise in the northern hemisphere and anticlockwise in the southern hemisphere.

Q 86.C

- **Kudremukh National Park** is a beautiful place, located in the Dakshina Kannada, Udupi and Chikmagalur districts of the state of Karnataka. Nestled in the Western Ghats. **Three important rivers, the Tunga, the Bhadra, and the Netravati are said to have their origin here.** The Tunga River and Bhadra River flow freely through the parklands. The animals found there include malabar civets, wild dogs, sloth bears and spotted deer. **Hence option (c) is the correct answer.**
- **Eravikulam National Park** is located along the Western Ghats in the Idukki district of Kerala in India. It is the first national park in Kerala. Eravikulam National Park is a UNESCO World Heritage Site. Anamudi, (2,695 meters), the highest peak in India south of the Himalayas is inside this park. Many perennial streams criss-cross the park. They merge to form tributaries of the Periyar river in the west and of the Cauvery River in the east.
- **Bengaluru Bannerghatta National Park** is located about 22 Kms south of the Bengaluru city of the state of Karnataka. It is one among the few places in the world where wilderness is preserved so close to a big city. The vision of the Bengaluru Bannerghatta National Park is to create an opportunity for the citizens - specially to children of Bangalore Metropolitan City to have a Biological Recreation Center very close to

the city in the midst of the forests in the valley of the famous Champakadhama hills inside the Bannerghatta National Park. To accomplish this vision, a mini zoo was created in 1972' which gradually grown into the present National Park by 2002.

- **Nameri National Park** is located in the foothills of the Eastern Himalayas in the Sonitpur District of Assam, India, about 40 kilometres from Tezpur. Nameri is a birder's paradise with over 300 species of birds. The area is crisscrossed by the river Jia-Bhoroli and its tributaries namely the Diji, Dinai, Doigurung, Nameri, Dikorai, Khari etc. There is a good prey base in the form of Sambar, Barking deer, Hog Deer, Wild Boar and Gaur. About 3000 domestic cattle also form part of this prey base for Tiger and Leopards.

Q 87.A

- The column of atmosphere is divided into five different layers depending upon the temperature condition. They are: troposphere, stratosphere, mesosphere, thermosphere and exosphere.
- **Statement 1 is correct:** The troposphere is the lowermost layer of the atmosphere. Its average height is 13 km and extends roughly to a height of 8 km near the poles and about 18 km at the equator. Thickness of the troposphere is greatest at the equator because heat is transported to great heights by strong convectional currents. This layer contains dust particles and water vapour. All changes in climate and weather take place in this layer. The temperature in this layer decreases at the rate of 1°C for every 165m of height. This is the most important layer for all biological activity.
- The zone separating the troposphere from stratosphere is known as the tropopause. The air temperature at the tropopause is about minus 80 degree C over the equator and about minus 45 degree C over the poles. The temperature here is nearly constant, and hence, it is called the tropopause. The stratosphere is found above the tropopause and extends up to a height of 50 km. One important feature of the stratosphere is that it contains the ozone layer. This layer absorbs ultra-violet radiation and shields life on the earth from intense, harmful form of energy.
- **Statement 2 is not correct:** The mesosphere lies above the stratosphere, which extends up to a height of 80 km. In this layer, once again, temperature starts decreasing with the increase in altitude and reaches up to minus 100°C at the height of 80 km. The upper limit of mesosphere is known as the mesopause.
- The thermosphere is the layer in the Earth's atmosphere directly above the mesosphere and below the exosphere. It contains electrically charged particles known as ions, and hence, it is a part of region known as the ionosphere (Ionosphere includes the thermosphere and parts of the mesosphere and exosphere which are ionized by solar radiation). Radio waves transmitted from the earth are reflected back to the earth by this layer. Temperature here starts increasing with height.
- The uppermost layer of the atmosphere above the thermosphere is known as the exosphere. This is the highest layer but very little is known about it. Whatever contents are there, these are extremely rarefied in this layer, and it gradually merges with the outer space. Although all layers of the atmosphere must be exercising influence on us, geographers are concerned with the first two layers of the atmosphere.

Q 88.B

- **Recently, a new study has warned that Permafrost peatlands — frozen, carbon-storing expanses of land — in Europe and western Siberia might be approaching their tipping point faster than expected.**
- Peatlands are a class of wetlands, which are ecosystems flooded with water. **Waterlogged conditions limit microbial decay of dead plant materials rich in carbon dioxide.** This prevents the reintroduction of the gas into the atmosphere. **Hence, statement 2 is correct.**
- Some peatlands are buried under frozen ground or permafrost and exist as permafrost peatlands. They are found in the northern parts of Alaska, Canada, Russia and parts of northern Europe.
- Peatlands, which occupy only **3 per cent of the global land surface.** **Hence, statement 1 is not correct.**
- They store twice as much carbon as all the world's forests.

Q 89.B

- Big Bang Theory or the expanding earth theory given by Edwin Hubble. According to it, in the beginning, all matters forming the universe existed in one place in form of a "tiny ball" with an unimaginable small volume, infinite temperature and infinite density. At the Big Bang, that took place 13.7 billion years before the present, the "tiny ball" exploded and thereafter the universe continues to expand. This is the most widely accepted theory. **Hence, option 1 is correct.**
- The Steady-state theory was first put forward in 1948 by British scientists Sir Hermann Bondi, Thomas Gold, and Sir Fred Hoyle as an alternative to the big-bang hypothesis. According to it, the universe is

always expanding but maintaining a constant average density, with matter being continuously created to form new stars and galaxies at the same rate that old ones become unobservable as a consequence of their increasing distance and velocity of recession. A steady-state universe has no beginning or end in time, and from any point within it the view on the grand scale—i.e., the average density and arrangement of galaxies—is the same. Galaxies of all possible ages are intermingled. **Hence, option 2 is correct.**

- The Big Splat Theory: It is related to the formation of Moon as a satellite to earth. According to it, a body of the size one to three times that of Mars collided into the Earth sometimes shortly after earth was formed. It blasted a large part of the earth into space. This continued to orbit around the earth and eventually formed the present moon around 4.44 billion years ago. **Hence, option 3 is not correct.**

Q 90.D

- Different types of earthquake waves travel in different manners. As they move or propagate, they cause vibration in the body of the rocks through which they pass.
- **Statement 1 is not correct:** Primary waves (P-waves) vibrate parallel to the direction of the wave. This exerts pressure on the material in the direction of the propagation. As a result, it creates density differences in the material leading to stretching and squeezing of the material.
- **Statement 2 is not correct:** Density difference is created by P-waves (see the above explanation). The direction of vibrations of secondary waves (S-waves) is perpendicular to the wave direction in the vertical plane, hence they create troughs and crests in the material.

Q 91.D

- The earth's surface is being continuously subjected to external forces induced basically by energy (sunlight). That means, the earth's surface is being continuously subjected to by external forces originating within the earth's atmosphere and by internal forces from within the earth. **The external forces are known as exogenic forces and the internal forces are known as endogenic forces.**
- **The exogenic processes derive their energy from atmosphere determined by the ultimate energy from the sun and also the gradients created by tectonic factors. While, the energy emanating from within the earth is the main force behind endogenic geomorphic processes.** This energy is mostly generated by radioactivity, rotational and tidal friction and primordial heat from the origin of the earth. This energy due to geothermal gradients and heat flow from within induces diastrophism and volcanism in the lithosphere. **Hence, statement 1 is not correct.**
- **The actions of exogenic forces result in wearing down (degradation) of relief/elevations and filling up (aggradation) of basins/ depressions, on the earth's surface.** The phenomenon of wearing down of relief variations of the surface of the earth through erosion is known as gradation. **The endogenic forces continuously elevate or build up parts of the earth's surface** and hence the exogenic processes fail to even out the relief variations of the surface of the earth. **Hence, statement 2 is not correct.**

Q 92.B

- To keep up with the rising demand for edible oils, the central government announced plans to boost domestic production of oilseeds. Finance Minister noted that a rationalized and comprehensive scheme to increase domestic production of oilseeds will be implemented.
- The proposed scheme is in response to the findings of the Economic Survey for 2021-22 that vegetable oil consumption in India is expected to remain high due to population growth and urbanization. As per the **OECD-FAO Agricultural Outlook 2021-2030**, India is projected to maintain a high per capita vegetable oil consumption growth of 2.6 percent per annum, reaching 14 kg per capita by 2030, necessitating a high import growth of 3.4 percent per annum.
- India currently relies on imports to satisfy 60 percent of its domestic demand. **Palm oils account for 60 percent of all imported edible oils. Hence, statement 2 is correct.**
- While edible oil production in the country increased from 6.1 million tonnes (mt) in 2015-16 to 7.9 mt in 2019-20, import increased from 14.9 mt in 2015-16 to 15.6 mt in 2018-19, **before falling to 13.4 mt in 2019-20. Thus, the imports of edible oils have not steadily increased in the past five years. Hence, statement 1 is not correct.**

Q 93.A

- **Western Disturbances**
 - They are shallow cyclonic depressions (weak temperate cyclones) originating over the eastern Mediterranean Sea and travelling eastwards across West Asia, Iran, Afghanistan and Pakistan before they reach the northwestern parts of India. **Hence option (c) is correct.**

- They are steered in India by Westerly Jet Streams. **Hence option (d) is correct.**
- On their way, the moisture content gets augmented from the Caspian Sea in the north and the Persian Gulf in the south. Although the amount of rainfall caused by them is meagre, it is highly beneficial for rabi crops. **Hence option (b) is correct.**
- It sustains the flow of water in the Himalayan rivers during the summer months.
- An increase in the prevailing night temperature generally indicates an advance in the arrival of these cyclones disturbances. **Hence option (a) is not correct.**

Q 94.A

- Some Famous Local Storms of Hot Weather Season in India are:
 - Mango Shower: Towards the end of summer, there are pre-monsoon showers which are a common phenomena in Kerala and coastal areas of Karnataka. Locally, they are known as mango showers since they help in the early ripening of mangoes. Blossom Shower: With this shower, coffee flowers blossom in Kerala and nearby areas. Nor Westers: These are dreaded evening thunderstorms in Bengal and Assam. Their notorious nature can be understood from the local nomenclature of 'Kalbaisakhi', a calamity of the month of Baisakh. These showers are useful for tea, jute and rice cultivation. In Assam, these storms are known as "Bardoli Chheerha". Loo: Hot, dry and oppressing winds blowing in the Northern plains from Punjab to Bihar with higher intensity between Delhi and Patna.
 - Mustard is a Rabi crop, therefore storms of Hot Weather Season do not assist in its growth. **Hence option (a) is the correct answer.**

Q 95.B

- Pacific Ocean Currents includes various cold and warm current which moves clockwise circulation in the Northern Pacific Ocean and Anticlockwise circulation in South Pacific Ocean thereby influencing the climatic pattern in the coastal regions. The currents are mentioned below:
- North Equatorial Current (Warm)
- South Equatorial Current (Warm)
- Counter Equatorial Current (Warm)
- **Kuroshio System (Warm)**
- North Pacific Drift (Warm)
- **Oyashio Current (Cold)**
- **California Current (Cold)**
- Peruvian or Humboldt Current (Cold)
- East Australia Current (Warm)
- Alaska current (warm)
- **Hence option (b) is the correct answer.**



Q 96.D

- As shown in the adjoining figure, all three are correct. A line of dots in the central parts of the Atlantic Ocean runs almost parallel to the coastlines. It further extends into the Indian Ocean. It bifurcates a little south of the Indian subcontinent with one branch moving into East Africa and the other meeting a similar line from Myanmar to New Guinea. This line of dots coincides with the midoceanic ridges. The shaded belt showing another area of concentration coincides with the Alpine-Himalayan system and the rim of the Pacific Ocean. The map of volcanoes also shows a similar pattern. The rim of the Pacific is also called rim of fire due to the existence of active volcanoes in this area. **Hence, answer is (d).**

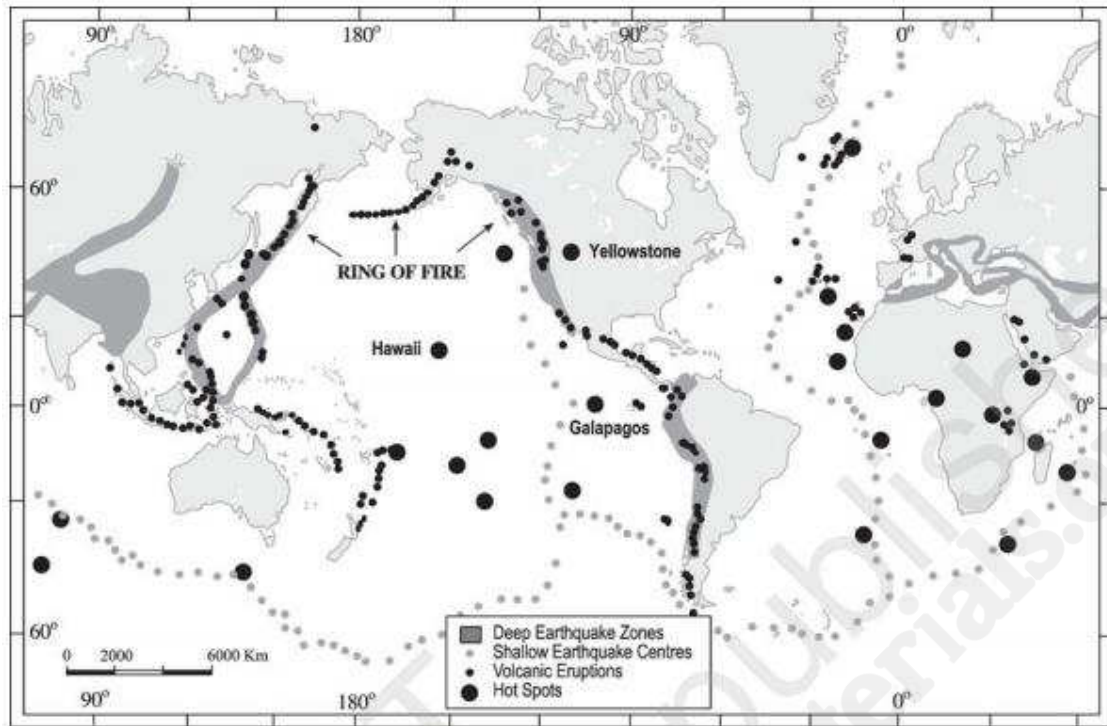


Figure 4. 2 : Distribution of earthquakes and volcanoes

Q 97.D

- From the Sub-Tropical High-Pressure Belts, winds blow towards the Temperate Low-Pressure Belts as the variable Westerlies. Under the effect of the Coriolis Force, they become the South-Westerlies in the northern hemisphere and the North-Westerlies in the southern hemisphere.
- They are more variable in the northern hemisphere. but they play a valuable role in carrying warm equatorial waters and winds to western coasts of temperate lands. This warming effect and other local pressure differences have resulted in a very variable climate in the temperate zones, dominated by the movements of cyclones and anticyclones.
- In the southern hemisphere where there is a large expanse of ocean, 40 degrees S to 60 degree S, Westerlies blow with much greater force and regularity throughout the year.** They bring much precipitation to the western coasts of continents. The weather is damp and stormy. **It is thus usual for seafarers to refer to the Westerlies as the Roaring Forties, Furious Fifties and Shrieking Sixties, according to the varying degree of storminess in the latitudes which they blow.**

Q 98.A

- Andhra Pradesh and West Bengal had recently withdrawn the “general consent” granted to the Central Bureau of Investigation (CBI), effectively curtailing the agency’s powers in the States without prior permission.
- Given that the CBI has jurisdiction only over central government departments and employees, it can investigate a case involving state government employees or a violent crime in a given state only after that state government gives its consent. Thus, it gets a general consent instead of a case-specific consent to avoid taking permission each time.**
- The general consent is normally given for periods ranging from six months to a year.
- Hence option (a) is the correct answer.**

Q 99.B

- **Statement 1 is not correct:** Longitudes is an angular distance, measured in degrees along the equator east or west of the Prime Meridian. This means one half of the world is measured in degrees of east longitude up to 180° , and the other half in degrees of west longitude up to 180° . Each line of longitude equals half of the circumference of the Earth because each extends from the North Pole to the South Pole and passes through the equator. Hence, all longitudes are of equal length irrespective of the location of the observer, whether one is at pole or equator.
- **Statement 2 is correct:** As the longitudes converge towards the poles, the distance between two corresponding longitudes decreases poleward. It is highest at the equator.
- **Statement 3 is not correct:** Altitude of place is its height of a place over sea level. It has no correlation with latitude or longitude of a place. Like Ural mountains in Russia have higher altitudes compared to Northern Indian plains and Mount Everest has altitude even higher than Ural mountains.

Q 100.B

- **Recently, the Land Port Authority of India celebrated its 10th foundation day.**
- **It was established as a statutory body under the Department of Border Management, Ministry of Home Affairs. Hence, statement 1 is not correct.**
- The Land Ports Authority of India is entrusted with the task of building land ports on India's borders and provide state of the art infrastructure facilities to facilitate **seamless and efficient movement of cargo and passengers** with an overall objective of reducing dwell time and trade transaction costs, promote regional trade and people-to-people contact and imbibe the best international practices. **Hence, statement 2 is correct.**